

NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

We develop Neural Bellman Operators that construct continuations from their own fitted futures and return feasible economic policies. A constructive backend relates primitive approximation resources to loss against the original Bellman optimum. For trained affine-ReLU continuations in the investment model, analytic innovation integration yields exact action witnesses. We prove that lossless interval screening preserves these witnesses, the resulting whole-cell certificates and a prospective controller's stopping decisions, while controlling the number of exact comparisons under explicit separation conditions. The fitted-witness transfer theorem keeps optimization, acquisition, continuation and verification errors distinct. In the original nonlinear investment family, non-tensor services in two, four and eight dimensions evaluate prespecified economic targets. A matched execution compares complete computation costs for identical returned policies; separate conventional-method comparisons retain their own policy-cost evidence. All failed targets, earlier applications and adverse outcomes are preserved.

KEYWORDS: Dynamic programming, Neural approximation, Policy certification, Information acquisition, Computational economics.

1. INTRODUCTION

A continuation approximation matters because it changes an economic decision. Its fitting error, the loss of its implemented policy, and the work needed to certify an improvement are different objects. This paper develops Neural Bellman Operators around that distinction. A construction uses its own fitted future values, returns feasible action witnesses, and connects numerical resources to the cost of the policy actually implemented.

There are two linked numerical contracts in the same controlled economy. Under effective primitive moduli, the constructive backend returns a feasible policy with a prescribed error account against the original Bellman optimum. Theorems 2 and 3 retain the construction and its exact continuous original-owner implementation. A fitted backend instead proposes actions from learned features. Those proposals become deployable only after a whole-cell policy certificate is satisfied. A prospective service then stops when the complete policy attains a prespecified expected-cost target relative to an installed controller. This additional contract neither changes the Bellman optimum nor turns an empirical fitting error into an optimality guarantee.

The trained neural backend has an explicit computational structure. Under the model’s scalar uniform innovation, the expectation of a one-hidden-layer ReLU continuation is analytic. For the scalar investment action, the integrated objective is piecewise quartic, and a finite collection of rationally isolated cubic stationary points supplies exact action-lattice witnesses. This produces a terminating action search for a fitted nonlinear continuation without invoking another unconverged nonconvex optimizer. It is distinct from compiling a native min-plus envelope into an identical neural circuit. The verification theorem nevertheless applies to both neural and conventional candidates.

The policy account is implemented on a locally split, non-tensor observation partition. A one-state residual and value cache encloses the old policy’s continuation without a full table of state pairs. A signed sensitivity calculation for the smooth installed reference further retains economically useful continuation direction. We prove its validity and give explicit state-dimension, leaf-count and horizon costs. The original nonlinear investment family has a dimension-uniform smooth-reference stability bound, although the work needed for a chosen approximation accuracy can still grow with dimension. Every later discontinuous acquired actor is handled by the generic cache, not by differentiating away its jumps.

The preceding prospective experiment compares trained ReLU, quadratic and tree-based Bellman continuations in dimensions two, four and eight and horizons two, four and six. The original compiled-witness and tensor-FVI frontends remain in the two-dimensional comparison. Each method pays for its own construction, verification, inference and durable output. Targets require two- or ten-percent

expected-cost reductions from installed zero investment. A separately frozen signed-sensitivity extension attains both targets in every declared service; its preceding cache-only failures remain in the record. The complete-policy costs, nonterminal changes and all-in clocks determine what this means economically. They do not establish uniform neural superiority: common verification actions often determine the same returned policy for several generators, and quadratic FVI is an important low-work comparator.

A separate sample-split calculation learns approximately null directions under uncertain action exposure. The guarantee qualifies both the nuisance correction and the remaining Bellman comparison with the same validation confidence box. It therefore concerns inferred structure, rather than merely a hand-selected exact invariant, while retaining the maintained nonlinear drift and innovation model.

The additional comparison in Section 13 executes the exact fitted action solver inside the prospective service, rather than assigning old cost observations to new policies. A nested-candidate design keeps the verifier and shared actions fixed, and genuinely different training seeds produce matched fitted objects for the menu and exact-search modes. Theorem 12 identifies how center optimization loss enters full-action Bellman accuracy; Proposition 14 supplies explicit action-sensitive acquisition and evaluation moduli. The experiment then separately asks whether the additional witness changes a certified decision and whether its returned policy has a lower actual economic cost.

1.0.0.1. Related methods. Approximate policy iteration and fitted value iteration have established approximation-error analyses (Munos, 2003, Munos and Szepesvári, 2008, Scherrer et al., 2015). Conservative and safe improvement relate a policy update to an incumbent (Kakade and Langford, 2002, Pirodda et al., 2013, Metelli et al., 2021). Tree-based fitted iteration provides a strong conventional comparator (Ernst et al., 2005). We do not identify these general principles as neural innovations. Our contribution is the connection among constructive and trained own-future continuations, original feasible witnesses, acquired-state implementation, computable directed verification, and a prospective economic return with complete work accounting. The original action-contrast development also distin-

guishes residual transport from state bisimulation metrics, reward transformations and value-equivalent model classes (Ferns, Panangaden, and Precup, 2004, Ng, Harada, and Russell, 1999, Grimm, Barreto, Singh, and Silver, 2020).

The article develops this constructive theorem–implementation–economic chain for expected costs. The complete development article and proof companion retain the broader recursive-preference, temporal-self, diffusion and game applications under their original hypotheses. All earlier experiments, including adverse comparisons, are preserved there and in the source archive. Linear action-kernel and sampling identities in the present chain are not silently extended to those different economic evaluations.

1.0.0.2. *Computing the same economic answer at lower recorded cost.* An exact trained-neural action witness also raises an implementation question. Analytically integrated ReLU features can be reduced exactly on an action region and used to eliminate nonminimizing lattice points through outward bounds. Section 9 proves that rational comparison of the survivors returns the original exact witness. It then proves that replacing the solver inside a mathematical prospective controller preserves its entire policy and stopping history. Section 14 tests this intervention with independently reconstructed, matched services in the original investment family. Its economic object is the complete cost of returning an identical policy, not a welfare difference between equivalent encodings. This extends the original construction–witness–verification–return chain while retaining the conventional comparisons and their actual cost classifications.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t , the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v , define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \quad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \quad (1)$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. The direct-cost inference in Section 10 is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1} , returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. Action-dependent errors and complete-policy loss

Let $H(a)$ be a true current cost and $\hat{H}(a)$ its continuation approximation. If $\hat{a}$ is δ -optimal for $\hat{H}$ on the same feasible action set, then

$$H(\hat{a}) - \inf_a H(a) \leq \delta + \text{osc}_a \{\hat{H}(a) - H(a)\}. \quad (2)$$

Indeed, compare $\hat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No

transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \leq u_t, \quad \mathcal{T}_t^\pi f_{t+1} - f_t \leq v_t, \quad \ell_T \leq f_T - g \leq u_T.$$

Then the actual feasible policy satisfies

$$0 \leq J_0^\pi - V_0^* \leq \sum_{t < T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \quad (3)$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

PROOF: Backward comparison and cash invariance give

$$V_t^* \geq f_t - \sum_{k=t}^{T-1} B_{t,k} u_k - B_{t,T} u_T, \quad J_t^\pi \leq f_t + \sum_{k=t}^{T-1} B_{t,k} v_k - B_{t,T} \ell_T,$$

where $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$. The terminal claim is the stipulated enclosure; applying the Bellman or fixed-policy operator proves each inductive step. Subtraction gives (3). A shift $f_t \mapsto f_t + c_t^0$ adds $c_t^0 - \beta_t c_{t+1}^0$ to u_t and subtracts it from v_t . The terminal width is unchanged. *Q.E.D.*

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \leq K_t^A \|x - z\|_1, \quad L_t = A_t^Q + D_t K_t^A, \quad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t -Lipschitz whenever f_{t+1} is L_{t+1} -Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t , form effective feasible k_t -nets. Evaluate their own-future objectives with certified error at most e_t . Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \leq K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \quad (4)$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min_i \arg \min \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g .

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine-ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \leq f_t - \mathcal{T}_t f_{t+1} \leq D_t k_t + e_t + 2L_t h_t, \quad (6)$$

$$\mathcal{T}_t^\pi f_{t+1} - f_t \leq e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \leq \sum_{t < T} B_t(D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T(2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

PROOF: The selected node label lies between the node Bellman infimum minus e_t and that infimum plus $D_t k_t + e_t$. The Lipschitz property of the infimum implies that

every cone lies above it minus e_t . At a nearest node its cone exceeds the infimum by at most $D_t k_t + e_t + 2L_t h_t$. At the attaining original index, the two feasible pairs and (4) give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) \leq y_{t,i} + e_t + L_t \|x - x_{t,i}\|_1 + D_t \zeta_t.$$

This proves (6). The terminal residual belongs to $[-e_T, e_T + 2L_g h_T]$; Proposition 1 gives the loss bound. A finite minimum of equally Lipschitz cones is equally Lipschitz. The identities $|z| = \text{ReLU}(z) + \text{ReLU}(-z)$ and $\min(u, v) = u - \text{ReLU}(u - v)$ give the exact network. Finite lexicographic comparison and Borel repair give measurability. Effective refinement of the finitely many positively weighted terms yields the termination assertion. Q.E.D.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. Original-owner compilation on a tensor cover

On a tensor cover $\mathcal{X} = \prod_{j=1}^d \{x_{j,0}, \dots, x_{j,n_j}\}$, put $S = \prod_j (n_j + 1)$. For arbitrary labels y_i , including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex}, i} (y_i + L \|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v .

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v \in \mathcal{V}(C)} (z_v + L \|x - v\|_1), \tag{8}$$

$$(f(x), i_*(x)) = \min_{\text{lex}, v \in \mathcal{V}(C)} (y_{o_v} + L \|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

PROOF: The triangle inequality bounds $f(x)$ above by each corner expression. Choose the lexicographically first attaining site $i_*(x)$. In every coordinate choose the cell endpoint between x_j and $x_{i_*,j}$. At this corner distances add exactly. An owner of strictly smaller transformed value would improve the objective at x ; a tied smaller owner would contradict the original tie rule. Thus this corner has owner $i_*(x)$, proving both identities. Along a coordinate line, forward and backward relaxations respectively minimize over sites on either side, always comparing value–original-index pairs. Iterating over coordinates gives the separable distance transform and its stated work. Rational arithmetic and unchanged repair complete the claim. *Q.E.D.*

The nodal algorithm is the classical distance transform (Felzenszwalb and Huttenlocher, 2012); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1} Q_t)$ in representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. Acquired states and conventional actors

A b -bit coordinate sensor reports a dyadic cell and its midpoint $\hat{x}$. On the unit cube, $\|x - \hat{x}\|_1 \leq r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t . This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\hat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error v_t .

Thus the implemented-policy account adds

$$\sum_{t < T} B_t \{ (2L_t + 2D_t K_t^A) r(b) + D_t \Delta_t + v_t \}. \quad (9)$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\text{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{ y_{i(x)} + L_t^{\text{gr}} \|x - x_{i(x)}\|_1 - f_t(x) \}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (48).

4. ALLOCATING RESOURCES TO AN ECONOMIC TOLERANCE

A single resolution parameter imposes an economic restriction that the Bellman problem does not impose. State coverage, action search and integration have different error coefficients and different effects on work. Refining all three at the same rate can therefore spend most of the service budget on the wrong source of error. We separate these choices while retaining the continuation, feasible-witness interpretation and all-state comparison developed above.

4.1. A separated primitive budget

Consider the investment family specified in Section 12.5, with $d \in \{2, 3, 4\}$ state coordinates, discount $\beta = 15/16$, and uniform coordinate grids with N intervals.

At a state node use $K + 1$ feasible action fractions. Integrate the common scalar innovation using M equal-probability bins and their midpoints. The three integers need not agree. Let L_t be the primitive graph modulus of the constructed witness continuation, and let D_t be its action modulus. A downward numerical displacement of the action grid is denoted by δ_a .

The state and action covering radii satisfy $h_t = d/(2N)$ and $k_t \leq 1/(8K) + \delta_a$. The innovation lies in an interval of length $1/16$. Within a bin its expected absolute distance from the midpoint is $1/(64M)$. Since changing the scalar innovation by $z - z'$ changes the next state by at most $d|z - z'|$ in the ℓ^1 norm, a certified numerical query has an error allowance

$$e_t \leq \frac{\beta d L_{t+1}}{64M} + \xi_t. \quad (10)$$

Here ξ_t covers arithmetic, interval evaluation and label rounding. It does not suppress the integration error. The terminal label allowance is e_T .

PROPOSITION 4—Separated-resource policy account: *Under the feasible-witness hypotheses, the returned policy satisfies*

$$0 \leq J^\pi - V^* \leq \frac{a}{N} + \frac{b}{K} + \frac{q}{M} + \eta, \quad (11)$$

where

$$\begin{aligned} a &= d \sum_{t < T} \beta^t L_t + d \beta^T L_g, & b &= \frac{1}{8} \sum_{t < T} \beta^t D_t, \\ q &= \frac{d}{32} \sum_{t < T} \beta^{t+1} L_{t+1}, & \eta &= \sum_{t < T} \beta^t (2\xi_t + D_t \delta_a) + 2\beta^T e_T. \end{aligned} \quad (12)$$

The primitive coefficients are independent of the computed labels. The leading number of innovation evaluations in target formation is

$$T(N + 1)^d (K + 1)M. \quad (13)$$

The compiled continuation additionally incurs its corner-evaluation and preprocessing costs. Pilot selection, policy extraction, verification, arithmetic bit lengths and output are separate charged components.

PROOF: Insert the state and action radii and (10) into the feasible-witness loss bound. The selected-policy upper account contributes the second copy of e_t , not a second action-search allowance. Collecting the coefficients of N^{-1}, K^{-1}, M^{-1} gives (12). There are $(N + 1)^d$ state nodes, $K + 1$ actions per node and M innovation representatives per action at each nonterminal date. No independence of fitting errors is used. Q.E.D.

The account explains why a common further state rung need not be a diagonal refinement. In the $T = 3, p = 4, d = 2$ cell, the exact primitive coefficients are approximately $a = 51.1569, b = 2.3650$ and $q = 1.0486$. The choice $(N, K, M) = (128, 16, 8)$ has account about $0.6786 + \eta$, below target one when the numerical remainder fits the remaining budget. The diagonal choice $(128, 128, 128)$ is more accurate, but its target-generation innovation count at that same state grid is $129 \cdot 128 / (17 \cdot 8)$ times larger. This is an arithmetic comparison of two declared allocations, not an observation of the unexecuted diagonal service or a runtime ratio. Both witness and conventional FVI receive the separated allocation in the experiment.

4.2. The optimal continuous allocation and its boundary cases

A useful planning problem minimizes the leading product $N^d KM$ subject to (11). This proxy isolates the primitive evaluation term; it is not the entire operation count or the measured wall clock. Write $E = \varepsilon - \eta > 0$ and suppose first that a, b, q are positive and all optimal resources exceed their required lower bounds.

PROPOSITION 5—Error shares and clipped allocation: *The unique interior minimizer of the continuous proxy has*

$$\frac{a}{N} = \frac{d}{d+2}E, \quad \frac{b}{K} = \frac{1}{d+2}E, \quad \frac{q}{M} = \frac{1}{d+2}E. \quad (14)$$

For lower resource bounds $r_i \geq \underline{r}_i > 0$, put $c = (a, b, q)$, $p = (d, 1, 1)$ and $\bar{x}_i = c_i / \underline{r}_i$. If $E < \sum_i \bar{x}_i$, there is a unique positive λ such that

$$x_i = \min\{\bar{x}_i, p_i \lambda\}, \quad \sum_i x_i = E; \quad r_i = c_i / x_i. \quad (15)$$

These resources minimize the constrained continuous proxy. If $E \geq \sum_i \bar{x}_i$, all lower resource bounds suffice. Coordinates with zero error coefficient remain at their resource lower bound. When dyadic lower bounds are used, rounding each optimal resource upward to a power of two preserves the error guarantee and multiplies this proxy by less than 2^{d+2} .

PROOF: Set $x_i = c_i/r_i$. Minimizing the resource product is equivalent to maximizing $\sum_i p_i \log x_i$ over $0 < x_i \leq \bar{x}_i$ and $\sum_i x_i \leq E$. Strict concavity and the multiplier conditions give p_i/x_i equal on unconstrained coordinates and larger on capped coordinates. This is precisely (15). The sum of its right sides is strictly increasing until all caps bind. The unconstrained case yields (14). Upward dyadic rounding never enlarges an error term and multiplies $N^d KM$ by less than the stated factor. Neither argument bounds rational-arithmetic bit cost or computer time. *Q.E.D.*

The optimization is a standard separable resource-allocation calculation. Its role here is to expose the error coefficients of the actual own-future policy construction, including integration and implementation, rather than to claim novelty for the multiplier argument. A finite experimental menu need not contain the continuous optimum. The prototype uses a common, frozen dyadic menu for all generators and records all of it. It does not describe this menu as a globally optimal tuning policy for either FVI or the witness method.

4.3. A complete tolerance frontier

For method m , let a prospectively fixed ladder return certificates $g_{m,r}$ and cumulative costs $w_{m,r}$, including failed earlier rungs. Define

$$r_m(\varepsilon) = \min\{r : g_{m,r} \leq \varepsilon\}, \quad W_m(\varepsilon) = w_{m,r_m(\varepsilon)}. \quad (16)$$

An empty set is a capped failure, not an interpolated success. Sort the finite union of all certificate values and zero. On each left-closed, right-open interval between successive values, every method's first crossing and complete work are constant. Evaluating the first-crossing rule at the left endpoint therefore reconstructs the entire positive-tolerance frontier exactly, even when successive certificates are not

monotone. An unbounded final interval is retained. Repetitions have separate costs, while their mathematical checkpoints must be checked for identity.

This finite partition supplies the appropriate interpretation of a threshold advantage. At a fixed resolution the witness method may cost more. At a tolerance where it crosses one rung earlier, the smaller certificate may more than offset that cost. Whether this happens, and over which tolerance interval, is determined by (16); it is not determined by a favorable single threshold. Under an additional homogeneous model $g_m(n) = A_m/n$ and $w_m(n) = C_m n^\kappa$, the continuous frontier ratio is $(C_W/C_F)(A_W/A_F)^\kappa$. This last expression is a conditional explanation, not a fitted scaling law for the experiments. The release supplies the exact finite partition, sensitivity around one, complete prefixes and separate fixed-resolution observations.

5. THE NONLINEAR INVESTMENT FAMILY

The economic law is the original dimension-normalized investment family. For $x \in [0, 1]^d$, scalar action $0 \leq a \leq \kappa(x)$, and cyclic coordinate $v = j + 1$,

$$\kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_j x_j, \quad (17)$$

$$F_j(x, a, Z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_v}{8} + \frac{x_j(1 - x_v)}{16} + \gamma_j a + s_j Z,$$

where (γ_j, s_j) alternates between $(1/2, 1)$ and $(1/4, -1)$ and Z is uniform on $[-1/32, 1/32]$. The initial state is uniform on the cube and $\beta = 15/16$. The stage and terminal costs are

$$c(x, a) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2 + a^2 + 4a^4, \quad (18)$$

$$g(x) = \frac{4}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2 \left(\frac{1}{2} - \frac{2}{d} \sum_j x_j \right)_+^2.$$

The original coefficients and continuous laws are unchanged. The domain is invariant: the drift lies in $[1/16, 11/16]$, $a \leq 1/4$, $\gamma_j \leq 1/2$, and the innovation magnitude

is at most $1/32$. The results are for this theoretical economy, not calibrated installation fees or estimated welfare magnitudes.

6. TRAINED NEURAL CONTINUATIONS AND EXACT INNOVATION INTEGRATION

The constructive backend supplies a feasible reference without relying on the success of nonconvex optimization. A trained backend addresses a different computational question: whether features fitted to the operator's own continuation targets produce useful candidate actions at a prescribed resource budget. These two backends are not identified with one another. In particular, fitting the hidden weights below is not compilation of the min-plus construction in Theorem 2.

Let the transition have the one-factor form

$$F_t(x, a, Z) = \mu_t(x, a) + s_t(x, a)Z, \quad Z \sim \text{Unif}[-r, r].$$

The deterministic map μ_t may be nonlinear and the state may be multidimensional. Consider a fitted continuation with m hidden units,

$$h(y) = c + v^\top y + \sum_{j=1}^m u_j (w_j^\top y + b_j)_+. \quad (19)$$

Both hidden and output coefficients may be data dependent. Once a candidate has been constructed, its stored coefficients specify an exact numerical function; uncertainty about whether training found a good function is not an arithmetic uncertainty about the function that was stored.

LEMMA 6—Integrated neural continuation: Put $z_j = w_j^\top \mu_t(x, a) + b_j$ and $R_j = r|w_j^\top s_t(x, a)|$. Then

$$P_t^a h(x) = c + v^\top \mu_t(x, a) + \sum_{j=1}^m u_j \Phi(z_j, R_j), \quad (20)$$

where $\Phi(z, 0) = z_+$ and, for $R > 0$,

$$\Phi(z, R) = \begin{cases} 0, & z \leq -R, \\ (z + R)^2 / (4R), & -R < z < R, \\ z, & z \geq R. \end{cases}$$

For fixed stored coefficients and evaluable primitives, a point expectation uses $O(md)$ arithmetic operations in addition to evaluating μ_t and s_t . Outward evaluation on a state-action box gives a valid enclosure. There is no innovation-grid discretization remainder in this analytic expectation.

PROOF: The scalar random variable $w_j^\top s_t Z$ is uniform on $[-R_j, R_j]$, or is identically zero. Integrating its positive part gives the displayed three cases. Linearity of expectation gives (20); the affine innovation term has mean zero. Each feature requires two inner products and a bounded number of scalar operations. The function Φ is nondecreasing in each argument on $R \geq 0$: in the interior its radius derivative is $(R^2 - z^2)/(4R^2) \geq 0$, and outside the interior it is constant in the radius. Endpoint evaluation with directed arithmetic, enclosing an uncertain branch rather than choosing a favorable branch, therefore preserves containment. Negative output weights must reverse interval endpoints in the usual way. *Q.E.D.*

The formula is a property of this neural class under the stated innovation law, not a general exact-integration result for arbitrary deep networks or unknown kernels. Quadratic critics also have analytic expectations here; tree critics use valid bin enclosures. Thus eliminating quadrature for the neural backend is a specific computational facility, not a claim that conventional approximation necessarily needs quadrature. On the active hinge region the equivalent expression $(z + R)^2/(4R)$ avoids subtracting two nearly equal squares. At $R = 0$ the separate branch is essential.

The fitted construction proceeds backward. At each date it draws training states, forms a finite feasible action menu, evaluates current cost plus the integrated *own* subsequent critic, minimizes these labels, and fits the next critic. Its training labels are not evaluations of the optimal Bellman continuation. Fitted value iteration with general function classes has a substantial literature; see [Munos and Szepesvári \(2008\)](#) and the tree-based construction of [Ernst et al. \(2005\)](#). Our use of training is accompanied by a separate whole-cell verification obligation. No training mean-square error is substituted for that obligation.

In the executed neural construction the hidden layer has width 32. All hidden and output coefficients receive 120 full-batch Adam updates, followed by a ridge fit of

the affine and output coefficients on the learned features. Coefficients are rounded to 24 binary fractional bits. The verifier evaluates this quantized function, not a hypothetical unrounded optimizer. The fitted hidden weights, targets, states, seeds and quantized output are retained. A fitted nonlinear network, a quadratic continuation, an ExtraTrees continuation, and the two original constructive frontends are therefore identifiable, distinct candidate generators. Whether their additional proposals change the deployed policy is an empirical question that we report separately from their fitting errors.

6.1. Certified action search for the fitted neural continuation

The same integrated representation also permits a constructive action search, rather than a second unconverged optimization problem. Fix a reported state in the original economy, so the deterministic next state is $\mu(x, 0) + \gamma a$. Let the feasible deployed actions be $\{0, \Delta, \dots, M\Delta\}$, with $M\Delta$ no greater than the whole-cell lower capacity. Interpret stored coefficients and the reported state as exact dyadic numbers.

THEOREM 7—Algebraic neural action witnesses: *For a continuation of the form (19), its integrated Bellman objective in the original scalar investment action has at most $2m$ interior breakpoints. Between consecutive breakpoints it is a quartic polynomial*

$$4a^4 + q_2a^2 + q_1a + q_0,$$

with rational coefficients. An exact minimizer on the deployed action lattice can be found among at most $22m + 13$ candidate indices, independently of the number of lattice points. Construction requires $O(md + m^2 + m \log m)$ rational arithmetic and comparison operations, plus $O(m)$ cubic-root isolation problems at the action precision. Coefficient bit lengths and root-isolation work must also be charged. The selected action is an original fitted-critic action witness, not a min-plus encoding.

PROOF: For feature j , write its preactivation mean as $z_j + s_j a$ and its innovation radius as R_j . When $s_j \neq 0$, branch changes occur only at $(\pm R_j - z_j)/s_j$; for $R_j = 0$ these coincide. Retain those breakpoints inside the feasible interval and its two endpoints. Lemma 6 makes each feature quadratic, affine or zero between breakpoints.

Adding $a^2 + 4a^4$ and the affine part of the critic gives the displayed polynomial. Its derivative is $16a^3 + 2q_2a + q_1$ and has at most three real roots in each of at most $2m + 1$ pieces.

Between stationary points and breakpoints the objective is monotone. Hence a lattice minimizer lies immediately to either side of a breakpoint or a stationary point, or at a feasible endpoint. Isolate each stationary root in a rational interval of width at most $\Delta/4$ and include every index from the floor of its lower endpoint divided by Δ through the ceiling of its upper endpoint divided by Δ . This contributes at most three indices per isolated root and contains the required neighboring lattice points. The breakpoints and endpoints contribute at most $2(2m + 2)$ indices. Thus $9(2m + 1) + 4m + 4 = 22m + 13$ is a valid bound. Extra indices and duplicate candidates do not invalidate minimization. Exact rational evaluation and lexicographic value-action comparison settle ties.

Computing feature coefficients costs $O(md)$, sorting the breakpoints costs $O(m \log m)$ comparisons, and literal formation of the polynomial on each piece costs $O(m^2)$ operations. Rational isolation of the finitely many cubic roots terminates at positive action precision, including multiple roots. This proves the construction and the stated separation between arithmetic counts and bit-dependent root-isolation work. *Q.E.D.*

The solver is implemented with rational root isolation and is checked against exhaustive exact lattice minimization, including saved fitted critics from each of the three state dimensions. It minimizes the stored critic's objective at the reported state. Whole-cell safety still requires the true-advantage certificate; critic optimality alone does not justify deployment. The preceding R55 economic experiment used its prospectively fixed 17-action construction menu. Its policies and measured clocks are not retroactively replaced by this new action solver, and its cost intervals are not attributed to untested replacement policies.

The new experiment in Section 13 executes the algebraic solver in independently reconstructed prospective services. Its policies, complete work and validation streams are recorded separately from that preceding experiment.

7. A COMPLETE-POLICY CERTIFICATE ON NON-TENSOR ACQUIRED STATES

For the remainder of the new verification and stopping results, the recursion is the bounded finite-horizon expected-cost model. Let $\mathcal{C}$ be a finite measurable partition, with closed boxes used only to verify the cells. Boundary ties follow a specified observation rule. A deployed action is constant on each reported cell and feasible at every state in its closure. A verification box meeting several cells must retain every possible future action. The argument does not require a tensor partition or a continuous actor.

Fix a complete incumbent π^k and define its true advantage

$$A_t^k(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^{\pi^k}(x) - J_t^{\pi^k}(x).$$

The incumbent has advantage zero. For each cell C , let a_i be robustly feasible candidates and let $\mathcal{B}_t(C)$ be a finite interval cover containing every action feasible at every $x \in C$. The cover may include additional actions; its bounds must remain valid on the feasible subset. Suppose verified endpoints satisfy

$$A_t^k(x, a_i) \leq u_i(C), \quad \ell_B(C) \leq A_t^k(x, a) \quad (a \in B \cap A_t(x)).$$

Set $U_t^k(C) = \min\{0, \min_i u_i(C)\}$ and $L_t^k(C) = \min\{0, \min_B \ell_B(C)\}$. Select the candidate attaining U , retaining the incumbent on a tie. All dates are computed against the same old policy and adopted simultaneously.

THEOREM 8—Directed policy improvement: *Under these conditions the new policy is feasible and*

$$J_t^{\pi^{k+1}}(x) \leq J_t^{\pi^k}(x) \tag{21}$$

for every state and date. With $E_t^k = \sup_x (J_t^{\pi^k}(x) - J_t^*(x))$ and $\epsilon_t^k = \max_C (U_t^k(C) - L_t^k(C))$,

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \epsilon_t^k. \tag{22}$$

For $B_{t,j} = \prod_{i=0}^{j-1} \beta_{t+i}$, $B_{t,0} = 1$, and $m = \min\{K, T - t\}$,

$$E_t^K \leq \sum_{j=0}^{m-1} B_{t,j} \epsilon_{t+j}^{K-1-j} + B_{t,m} E_{t+m}^{K-m}.$$

The final term vanishes when $K \geq T - t$. Exact full-action greedy certificates thus recover the original Bellman optimum within T passes. The coefficient one multiplying the directed error cannot be reduced uniformly.

PROOF: The selected advantage is at most $U \leq 0$. Starting with the common terminal value, backward monotonicity proves (21). The full-action lower cover gives

$$A_t^k(x, \pi_t^{k+1}(x)) - \inf_{a \in A_t(x)} A_t^k(x, a) \leq U_t^k(C) - L_t^k(C).$$

Replacing the new continuation by the old one using safety, and comparing with an arbitrarily near-optimal Bellman action, yields (22). Taking the infimum limit avoids an attainment assumption. Iterating the inequality gives the displayed finite-horizon expansion. For sharpness take one decision date, an incumbent costing $\delta > 0$, and a feasible unproposed action costing zero. An exact incumbent upper endpoint is zero and an exact covering lower endpoint is $-\delta$. Retaining the incumbent gives both error and directed allowance equal to δ . *Q.E.D.*

Theorem 8 restates the original directed result in the partition notation used by the new implementation. It is deliberately representation neutral. Its premise is stronger than checking a finite candidate menu: without the full-action cover, a safe improvement need not have a small gap from the continuous-action optimum. A change of acquisition partition between passes is permitted when the old policy is transferred exactly to its child cells; the complete incumbent remains the same function.

7.1. A one-state residual and value cache

Let $h_T = g$ and h_t be any stored critics, including fitted networks. Write $e_t = J_t^\pi - h_t$ and

$$r_t = c_t(\cdot, \pi_t) + \beta_t P_t^{\pi_t} h_{t+1} - h_t, \quad e_t = r_t + \beta_t P_t^{\pi_t} e_{t+1}.$$

A cellwise residual interval $\mathcal{R}_t(C)$ encloses r_t throughout the cell. If $\mathcal{E}_{t+1}(C')$ encloses the next error, each innovation bin maps a current box and action into a

next-state box. The hull of $\mathcal{E}_{t+1}(C')$ over all cells intersecting that box encloses every possible next error. Denote the probability-weighted sum of those hulls by $\mathfrak{P}_t^a \mathcal{E}_{t+1}(C)$. Bin probabilities refer to the original continuous law, not an empirical law of midpoints.

PROPOSITION 9—Cached residual verification: *Starting from $\mathcal{E}_T = \{0\}$, the backward recursion*

$$\mathcal{E}_t(C) = \mathcal{R}_t(C) + \beta_t \mathfrak{P}_t^{\pi_t} \mathcal{E}_{t+1}(C) \quad (23)$$

encloses e_t on every cell. An independently computed direct value interval $\mathcal{V}_t(C)$ may be intersected with $h_t(C) + \mathcal{E}_t(C)$, and $\mathcal{E}_t(C)$ with $\mathcal{V}_t(C) - h_t(C)$. These intersections preserve containment. For feasible a, b the sum

$$c_t(C, a) - c_t(C, b) + \beta_t \{(P_t^a - P_t^b)h_{t+1}(C) + \mathfrak{P}_t^a \mathcal{E}_{t+1}(C) - \mathfrak{P}_t^b \mathcal{E}_{t+1}(C)\}$$

encloses the fixed-incumbent action difference. Intersecting it with the analogous direct-value difference gives a valid directed interval. Identical actions have the exact interval $\{0\}$.

PROOF: For every innovation in a bin, the next state belongs to one of the intersected cells; its error lies in the corresponding stored interval and hence in their hull. Integration against the true bin probabilities preserves order. Backward induction in the residual identity proves (23). Each displayed intersection combines two sets known to contain the same real value. Substituting $J^\pi = h + e$ in the action difference proves the final statement. An empty intersection indicates an invalid numerical calculation and must stop acceptance, rather than licensing a selected midpoint. Q.E.D.

This construction stores one-state bands rather than a table over state pairs. It sacrifices dependence between two trajectories when taking separate hulls; in particular, actor ambiguity and broad residual ranges can make it conservative. For locally Lipschitz critics, a centered residual evaluation can be intersected with the natural interval. A valid generalized-gradient enclosure suffices for the line-segment mean-value bound of a ReLU critic. This is not permission to differentiate

a discontinuous acquired actor. On one cell its action is fixed; across cells the cache explicitly retains the alternative actions.

7.2. Signed sensitivity of the installed reference

The original investment family in Section 5 has a smooth zero-investment reference. Its stage and terminal state costs are continuously differentiable, including the squared shortage term. Let μ denote the zero-noise drift and γ the action exposure. For a state box X_0 , propagate boxes X_{j+1} containing $\mu(X_j, 0) + sZ$ for the complete innovation support. Starting from a terminal gradient enclosure $\mathcal{G}_r \supset \nabla g(X_r)$, put

$$\mathcal{G}_j \supset \nabla c(X_j, 0) + \beta D_x \mu(X_j, 0)^\top \mathcal{G}_{j+1}. \quad (24)$$

The inclusion means outward interval multiplication and addition. It is evaluated backward along forward state tubes, with the true sparse Jacobian.

PROPOSITION 10—Signed reference certificate: *The interval $\mathcal{G}_0$ encloses ∇J_0^0 throughout X_0 . For a nonnegative proposed action a , enclose the gradient of the remaining zero policy at every state $\mu(x, \theta a) + sZ$, $0 \leq \theta \leq 1$. If $[v_-, v_+]$ encloses the average projection $\mathbb{E}[\gamma^\top \nabla J_{t+1}^0]$ over that segment, then*

$$A_t^0(x, a) \in a^2(1 + 4a^2) + \beta a[v_-, v_+]. \quad (25)$$

The signed interval may be intersected with Proposition 9. The construction is valid for the complete zero reference; it does not differentiate a subsequently changed, cellwise constant policy.

PROOF: Differentiate the finite fixed-policy recursion. On the invariant compact domain, the derivatives are bounded, so differentiation and the finite conditional expectations may be interchanged. For every shock path the chain rule has the enclosed Jacobian form in (24); backward induction and expectation preserve the enclosure. The fundamental theorem of calculus along the first action segment gives

$$(P_t^a - P_t^0)J_{t+1}^0(x) = a \int_0^1 \mathbb{E}[\gamma^\top \nabla J_{t+1}^0(\mu(x, \theta a) + sZ)] d\theta.$$

The state-only current costs cancel and the remaining current difference is $a^2 + 4a^4$. This proves (25). Partitioning the first innovation into bins and integrating valid projected gradient hulls with their exact probabilities preserves the same argument. Q.E.D.

PROPOSITION 11—Reference stability and computational work: *In the dimension-normalized investment family, for every $d \geq 2$,*

$$\|D_x \mu\|_\infty \leq \kappa := 11/16, \quad \|\nabla c(\cdot, 0)\|_1 \leq 15/2, \quad \|\nabla g\|_1 \leq 10.$$

Consequently, with r stages remaining,

$$\text{Lip}_\infty(J_r^0) \leq \frac{15}{2} \sum_{j=0}^{r-1} (\beta\kappa)^j + 10(\beta\kappa)^r \leq \frac{1920}{91}, \quad \beta = 15/16. \quad (26)$$

For a partition with N leaves, A candidate and covering intervals per leaf, and q innovation bins, the literal cached range-query implementation uses

$$O(TAqN^2d)$$

cell-membership work, in addition to primitive and critic evaluation. Signed reference tubes over a full sweep use $O(ANqdT^2)$ work and can stream their innovation children. Their working state storage is $O(NdT)$ in addition to stored critics, cached bands and retained $O(TAN)$ endpoint records. No q^T recursion or full state-pair table is required.

PROOF: In row j , the only state derivatives of the drift are $1/2 + (1 - x_{j+1})/16$ and $1/8 - x_j/16$, both nonnegative. Their sum is at most $11/16$. The target, cyclic imbalance and shortage terms have gradient ℓ^1 bounds $5/2$, 1 and 4 for current cost; the terminal target bound is 5 . This gives the stated constants. The chain rule yields $L_r \leq 15/2 + \beta\kappa L_{r-1}$ with $L_0 \leq 10$. Its geometric expansion and $\beta\kappa = 165/256$ prove (26).

A queried next-state box is tested against N leaves in d coordinates. There are $O(TANq)$ such root-bin queries. The backward cache therefore has the displayed membership cost. Each gradient tube with r remaining dates visits $O(Nr)$ state rows and $O(Ndr)$ Jacobian coordinates for one first-innovation bin. Summing r

across current dates gives $O(ANqdT^2)$. Streaming bins requires storage for one forward tube and its backward gradients, not all future shock combinations. *Q.E.D.*

The stability constant is a bound for this smooth reference, not a dimension-free complexity theorem for solving arbitrary dynamic programs. Resolving a desired policy or all-state accuracy may require the leaf count, network width and action cover to grow. The membership bound explicitly charges a quadratic leaf-count dependence. The executed locally split partition removes a tensor-product representation; it does not prove that a fixed 32-leaf partition resolves every eight-dimensional economic policy.

7.3. What a tighter interval establishes

For fixed candidates, let $C_t(C)$ be the greater of $L_t(C)$ and the minimum of their lower endpoints, including the incumbent's zero. Then

$$U - L = (U - C) + (C - L), \quad U - C \geq 0, \quad C - L \geq 0. \quad (27)$$

The first term measures uncertainty remaining between candidate lower information and the selected upper bound; the second measures the additional continuous-cover allowance. Their maxima need not add to the maximum of their sum. Actor ambiguity, acquisition width, innovation enclosures and arithmetic can enter both terms, so these are not separately estimated causal error components.

More detailed refinement can be recorded without such a causal interpretation. Keep the candidate actions fixed, replace each candidate upper bound by the minimum of its old and refined bounds, and replace the global lower bound by the maximum of its old and refined bounds. Then $U_r \leq U_{r-1}$, $L_r \geq L_{r-1}$, and the certificate improvement at step r is exactly

$$(U_{r-1} - L_{r-1}) - (U_r - L_r) = (U_{r-1} - U_r) + (L_r - L_{r-1}) \geq 0.$$

State subdivision, action-cover subdivision, innovation refinement and higher-precision arithmetic can each supply an independent valid refinement. The present execution records the cache-to-signed-tube refinement and both terms of (27); it does not label an unexecuted six-way refinement experiment as measured evidence.

8. FROM FITTED ACTION WITNESSES TO ECONOMIC POLICY ACCURACY

An exact action witness has a definite but limited meaning: it removes optimization error for the stored continuation at a reported state. The next result identifies the further obligations needed to turn that statement into an error bound for the implemented economic policy. It also isolates a comparison that can be made without attributing the common verifier's contribution to a particular representation.

Fix a complete reference policy π , a date t , and an acquired cell C with representative x_C . Write

$$Q^h(x, a) = c_t(x, a) + \beta_t P_t^a h_{t+1}(x), \quad Q^\pi(x, a) = c_t(x, a) + \beta_t P_t^a J_{t+1}^\pi(x).$$

The superscripts refer to the continuation used, not to different economic primitives. Let G_C be a finite action lattice feasible throughout C , containing the reference action. The reference is assumed implementable on the cell; all selected actions are measurable under the same observation contract. For $g_C \in G_C$, define its reported-state optimization loss by

$$\kappa_C = Q^h(x_C, g_C) - \min_{b \in G_C} Q^h(x_C, b). \quad (28)$$

This is a nonnegative, finite quantity. Theorem 7 makes it exactly zero for its returned lattice witness, regardless of the training loss or the signs of the learned output weights.

For all $x \in C$ and $a, b \in G_C$, suppose the following nonnegative bounds are available:

$$|[Q^h(x, a) - Q^h(x, b)] - [Q^h(x_C, a) - Q^h(x_C, b)]| \leq \eta_C, \quad (29)$$

$$|[Q^\pi - Q^h](x, a) - [Q^\pi - Q^h](x, b)| \leq \chi_C. \quad (30)$$

Suppose also that every $a \in A_t(x)$ has a $b \in G_C$ with

$$Q^\pi(x, b) \leq Q^\pi(x, a) + \delta_C. \quad (31)$$

These are distinct requirements. The first concerns acquisition within a cell, the second the action-relevant continuation error, and the third the difference between the robust deployed lattice and the original state-dependent continuous feasible set.

Let $[l_C(g_C), u_C(g_C)]$ enclose the true reference advantage $Q^\pi(x, g_C) - J_t^\pi(x)$ throughout C . Let w_C be a verified upper bound on this interval's width. The adoption rule of Theorem 8 is applied to a candidate set containing g_C and the reference, with selected upper endpoint $U_C \leq \min\{0, u_C(g_C)\}$. Denote its complete-action lower endpoint by L_C .

THEOREM 12—Fitted witness transfer: *Under (28)–(31) and the valid whole-cell endpoints above, the selected action a_C satisfies, at every $x \in C$,*

$$Q^\pi(x, a_C) \leq J_t^\pi(x), \quad (32)$$

$$Q^\pi(x, a_C) - \inf_{a \in A_t(x)} Q^\pi(x, a) \leq \min\{U_C - L_C, \rho_C\}, \quad (33)$$

$$\rho_C := \kappa_C + \eta_C + \chi_C + \delta_C + w_C.$$

Consequently the directed finite-sweep recursion remains valid with

$$\varepsilon_t^k = \sup_C \min\{U_t^k(C) - L_t^k(C), \rho_t^k(C)\}.$$

In particular, exact neural action search removes κ_C from this bound, not the other four terms. The coefficient one on the sum ρ_C is sharp without additional restrictions.

PROOF: Fix $x \in C$, an arbitrary feasible action a , and its covering lattice point b . Equations (28)–(30) imply

$$Q^\pi(x, g_C) - Q^\pi(x, b) \leq \kappa_C + \eta_C + \chi_C.$$

Adding (31) gives a bound against $Q^\pi(x, a)$ with the additional term δ_C . The endpoint rule gives

$$Q^\pi(x, a_C) - J_t^\pi(x) \leq U_C \leq u_C(g_C) \leq Q^\pi(x, g_C) - J_t^\pi(x) + w_C.$$

Combining these inequalities and taking the infimum over a proves the ρ_C bound. The full-action lower cover gives the independent $U_C - L_C$ bound. Also $U_C \leq 0$, proving (32). Backward induction from the common terminal value gives safety for simultaneous adoption of all dates. The proof of Theorem 8 then applies with the

smaller valid local error; no optimizer attainment for the continuous infimum is needed.

For sharpness, take one decision date, one observation cell containing one state, exact fitted values, and two actions with costs zero and $\delta > 0$. Let the incumbent and only proposed action be the action of cost δ , but let the lattice and full lower cover contain both actions. Then $\kappa_C = \delta$, the other four terms are zero, and the retained policy has loss exactly δ . Thus the coefficient multiplying the unrestricted sum cannot be smaller than one. *Q.E.D.*

The width term records the conservatism of certifying the proposed action and then choosing among upper endpoints. It is necessary: the action with the smallest upper bound need not have the smallest true advantage. A separately verified bound on $u_C(g_C) - [Q^\pi(x, g_C) - J_t^\pi(x)]$ can replace the full interval width. Likewise, sharper contrast bounds may replace (29) and (30). Their validity must be established before they are used in the policy account.

8.1. A primitive continuous-action allowance

The continuous-action condition is explicit in the investment family. Let $\bar{a}(x)$ be its capacity, Δ the action spacing, and

$$m_C \Delta = \Delta \left\lfloor \frac{\inf_{x \in C} \bar{a}(x)}{\Delta} \right\rfloor, \quad G_C = \{0, \Delta, \dots, m_C \Delta\}.$$

If $Q^\pi(x, \cdot)$ is uniformly L_a -Lipschitz, then

$$\delta_C = L_a \{\text{osc}_C \bar{a} + \Delta\} \quad (34)$$

is admissible. For the smooth installed zero policy in the original normalized family, one may take

$$L_a = \frac{3}{4} + \frac{900}{91}. \quad (35)$$

PROOF: For a feasible a , use the smaller of its downward lattice rounding and $m_C \Delta$. Its distance from a is at most $\text{osc}_C \bar{a} + \Delta$, proving (34). The derivative of

$a^2 + 4a^4$ has absolute value at most $3/4$ on $[0, 1/4]$. Proposition 11 bounds the zero-policy value's Lipschitz constant in the maximum norm by $1920/91$. The action exposure has maximum norm $1/2$ and the discount is $15/16$. Therefore the continuation's action Lipschitz constant is at most $900/91$, proving (35). *Q.E.D.*

The primitive allowance is valid but can be conservative. The executed policy bound uses the direct continuous-action lower cover, not a claim that the primitive sum is always tighter. The transfer theorem permits both accounts and shows exactly where an improved action search can enter the original Bellman-accuracy objective.

8.2. Nested proposals and identifiable verification gains

Consider two candidate sets $M_0 \subseteq M_1$ for the same reference, acquired cells, complete-action lower cover and valid upper-endpoint procedure. The endpoints of shared actions must be identical; this condition prevents changes in a fitted residual cache from being confused with changes in the proposed actions. Both sets retain the reference with exact upper advantage zero.

COROLLARY 13—Nested-candidate certificate: *Let $U_i(C)$ be the minimum candidate upper endpoint for M_i . Then*

$$U_1(C) \leq U_0(C) \leq 0, \quad 0 \leq U_1(C) - L(C) \leq U_0(C) - L(C).$$

The improvement is strict precisely when an added candidate has upper endpoint strictly below $U_0(C)$. Both returned complete policies are statewise safe relative to the reference. These inequalities alone do not order their actual expected costs.

PROOF: Taking a minimum over a larger finite set proves the first inequalities. The valid full-action lower bound is no greater than every feasible action's advantage and hence no greater than either selected upper bound. This proves the nonnegative gap inequalities and the strictness criterion. Safety follows from Theorem 8. To see the last distinction, consider a single state and date, an incumbent of cost 2, an old candidate of cost 1 with advantage enclosure $[-1, -1/2]$, and a

new candidate of cost $5/4$ with exact advantage $-3/4$. The new candidate has the smaller upper endpoint and is adopted, although its cost is higher than that of the old candidate. Both improve the incumbent. A lower cover of -1 makes both certificates valid. *Q.E.D.*

In the new experiment the ReLU-exact mode retains every ReLU-menu action and adds the algebraic witness. Both use the same fitted coefficients at a given construction attempt and a critic-independent smooth-reference verifier. Thus the corollary is an observed, testable comparison rather than a representation claim. Exact center-objective regret, stricter upper endpoints, changed deployed actions and independent expected-cost differences are reported separately.

8.3. *A paid decision between returned policies*

Let $[l, u]$ cover $J(\pi_1) - J(\pi_0)$ on a simultaneous event after both policies have been selected. If adopting π_1 rather than π_0 incurs an additional replacement charge τ and measured computational work difference ΔW , its net cost difference at computation price $\lambda \geq 0$ lies in

$$[l + \tau + \lambda\Delta W, u + \tau + \lambda\Delta W]. \quad (36)$$

A negative upper endpoint supports adoption; a positive lower endpoint rules it out on the same event. Between those regions the comparison is unresolved. This affine transformation is valid for every declared nonnegative price on that event. It does not turn a normalized benchmark into an estimated installation fee. It links a particular fitted-action computation to the same economic policy and resource question as the original NBO construction.

8.4. *Explicit moduli for the trained neural representation*

The transfer bound need not leave acquisition error as an unspecified constant. In the original investment family, write the learned continuation as

$$h(y) = v^\top y + c + \sum_{j=1}^m u_j \sigma(w_j^\top y + b_j).$$

Let $s_j = w_j^\top \gamma$, let R_j be the feature's scalar-innovation radius from Lemma 6, and set $D_C = \text{diam } G_C$ and $r_C = \sup_{x \in C} \|x - x_C\|_\infty$. Define

$$d_j(D) = \begin{cases} 0, & s_j = 0, \\ \min\{1, |s_j|D/(2R_j)\}, & s_j \neq 0, R_j > 0, \\ 1, & s_j \neq 0, R_j = 0. \end{cases}$$

All these quantities are determined by the stored coefficients and the declared primitive law.

PROPOSITION 14—Action-sensitive neural moduli: *For the original separable current action cost and drift with maximum-norm Lipschitz constant $11/16$, the acquisition allowance*

$$\eta_C = \beta_t \frac{11}{16} r_C \sum_{j=1}^m |u_j| \|w_j\|_1 d_j(D_C) \quad (37)$$

satisfies (29). In particular, the affine part and every exactly action-null ridge contribute zero to this allowance. For the installed zero reference, a valid continuation-error allowance is

$$\chi_C = \beta_t D_C \left\{ \frac{960}{91} + |v^\top \gamma| + \sum_{j=1}^m |u_j s_j| \right\}. \quad (38)$$

Either allowance may be replaced by the minimum of itself and any independently valid sharper enclosure. Computing the displayed bounds requires $O(md)$ arithmetic, with outward rounding when evaluated numerically.

PROOF: Write the integrated ridge as $\Phi_R(z) = \mathbb{E}\sigma(z + RZ)$ for Z uniform on $[-1, 1]$. For $R > 0$, its derivative is zero, then affine with slope $1/(2R)$, then one. Hence the function

$$z \mapsto \Phi_R(z + s_j a) - \Phi_R(z + s_j b)$$

is Lipschitz with constant at most $\min\{1, |s_j(a - b)|/(2R)\}$. For $R = 0$ the difference of the two hinge derivatives is bounded in absolute value by one almost everywhere, giving the same conclusion with the declared convention. For $s_j = 0$ the function is identically zero. The preactivation mean varies by at most $\|w_j\|_1 (11/16) \|x - x_C\|_\infty$.

The affine continuation contrast is $v^\top \gamma(a - b)$, which does not depend on x . The state-only current cost cancels, and the current action-cost difference also does not depend on x . Summing the ridge contributions and discounting proves (37).

Under a common innovation, the zero-policy continuation contrast is bounded by $(1920/91)\|\gamma\|_\infty|a - b| = (960/91)|a - b|$. The one-Lipschitz hinge gives a fitted continuation contrast bound $\{|v^\top \gamma| + \sum_j |u_j s_j|\}|a - b|$. The triangle inequality and discount give (38). Independently valid upper bounds may be minimized because each bounds the same nonnegative quantity. The arithmetic count follows from computing the feature projections and norms. *Q.E.D.*

These formulas connect feature geometry to acquired-state accuracy without assuming that training discovers a null direction. They complement the confidence-qualified learned-direction result: estimated primitive coefficients must still range over the corresponding confidence set. In the present known-law action-search experiment, the direct signed reference enclosure determines adoption and the full-action lower cover determines the displayed Bellman error. The primitive moduli supply an additional certified account, not a retroactive claim that their conservative sum was the stopping criterion.

9. LOSSLESS SEARCH FOR TRAINED NEURAL ACTION WITNESSES

The fitted-witness transfer result separates an optimization error from the errors in acquisition, continuation evaluation and full-action verification. An exact solver eliminates the first error, but exactness alone does not make the solver economical. We now exploit the learned features to reduce the work of returning precisely the same action witness. The continuation, economic primitives and original Bellman-accuracy target remain unchanged.

9.1. *Exact reduction of integrated neural features*

Fix a state and a stored affine-ReLU continuation. Interpret its finite binary coefficients as rational numbers. With a scalar uniform innovation, the expectation

of a ridge is

$$\phi_r(z) = \begin{cases} ((z+r)_+^2 - (z-r)_+^2)/(4r), & r > 0, \\ z_+, & r = 0. \end{cases}$$

The radius r is the absolute projection of the innovation support on the ridge direction. After removing a constant independent of the action, the fitted objective in the investment model takes the form

$$F(a) = q_1 a + q_2 a^2 + 4a^4 + \sum_{j=1}^m \omega_j \phi_{r_j}(z_j + s_j a). \quad (39)$$

All coefficients are rational. Negative fitted weights are permitted. No convexity of the fitted continuation is assumed. The robust feasible action lattice is $\mathcal{L} = \{k\delta : 0 \leq k \leq K\}$, where δ and $K\delta$ are fixed by the acquisition contract.

A feature can be simplified before any lattice search. Let $[v_j^-, v_j^+]$ be the exact range of $z_j + s_j a$ on $[0, K\delta]$. If $v_j^+ \leq -r_j$, its contribution vanishes. If $v_j^- \geq r_j$, it is affine. If $r_j > 0$ and $[v_j^-, v_j^+] \subseteq [-r_j, r_j]$, it is quadratic. Absorb these contributions into a rational constant c_0 and the coefficients of a and a^2 . Only features crossing an activation-region boundary remain in the ridge sum. Consequently

$$F(a) = c_0 + \tilde{q}_1 a + \tilde{q}_2 a^2 + 4a^4 + \sum_{j \in \mathcal{C}} \omega_j \phi_{r_j}(z_j + s_j a) \quad (0 \leq a \leq K\delta). \quad (40)$$

This is an identity, not pruning by a small fitted weight. A feature with a large weight may be eliminated exactly, whereas a small crossing feature remains unless its whole contribution is enclosed.

9.2. An interval screen with an exact return

Write $f_k = F(k\delta) - c_0$. Compute outward enclosures $[\ell_k, u_k]$ of all f_k and put

$$b = \min_{0 \leq k \leq K} u_k, \quad S = \{k : \ell_k \leq b\}. \quad (41)$$

Compare the original objective exactly on S , choosing the smallest index on a tie. The weak inequality in the definition of S is essential. Floating-point endpoints reject a point only when they prove that it cannot minimize the original objective; they never supply an approximate answer in place of the exact comparison.

THEOREM 15—Lossless neural action search: Suppose the intervals in (41) are finite, ordered and valid. Then S contains every exact lattice minimizer. Exact rational comparison on S , with the same smallest-index convention, returns exactly the witness of exhaustive lattice minimization and of the terminating algebraic solver.

Let $\eta = \max_k(u_k - \ell_k)$. Every survivor satisfies

$$F(k\delta) - \min_{a \in \mathcal{L}} F(a) \leq 2\eta. \quad (42)$$

In particular, a unique minimizer is the only survivor whenever its gap to the second-best lattice point exceeds 2η .

More generally, let $\mathcal{M} \subseteq \mathcal{L}$ be the exact minimizer set. If, for some $\mu > 0$,

$$F(a) - \min_{\mathcal{L}} F \geq \frac{\mu}{2} \text{dist}(a, \mathcal{M})^2 \quad (a \in \mathcal{L}),$$

then

$$|S| \leq \min \left\{ K + 1, |\mathcal{M}| \left(1 + 2 \left\lfloor \frac{2\sqrt{\eta/\mu}}{\delta} \right\rfloor \right) \right\}.$$

The growth condition is an additional assumption, not a property inferred from a fitted ReLU representation.

PROOF: Let k_* minimize f_k . For every j , $f_{k_*} \leq f_j \leq u_j$, hence $\ell_{k_*} \leq f_{k_*} \leq b$ and $k_* \in S$. This argument retains all minimizers, including ties. Exact comparison with the same tie rule therefore returns the same index. The algebraic solver enumerates lattice neighbors of every activation boundary and isolated stationary point on each polynomial piece; its established exhaustive-search equivalence then gives the same witness.

For a survivor k , validity and the width bound imply

$$f_k \leq \ell_k + \eta \leq b + \eta \leq u_{k_*} + \eta \leq f_{k_*} + 2\eta.$$

This proves (42) and the unique-gap statement. Under the additional growth condition a survivor is within $2\sqrt{\eta/\mu}$ of some element of $\mathcal{M}$. There are at most $1 + 2\lfloor 2\sqrt{\eta/\mu}/\delta \rfloor$ lattice points within that distance of one lattice minimizer. Taking a union, and then the total lattice cardinality, proves the last bound. *Q.E.D.*

The generic screening implication uses only interval order. The implementation contribution here is the exact reduction of integrated trained ridges, outward evaluation of their reduced objective, and exact recovery of the original NBO witness. On a failed arithmetic check or a lattice exceeding the declared screening limit, the implementation calls the original terminating algebraic solver. A fallback is charged as work and cannot be recorded as an approximate successful screen. A wide but valid enclosure may retain the entire lattice; the theorem promises correctness in that case, not a speed gain.

PROPOSITION 16—Arithmetic work and working storage: *Let $N = K + 1$, let $m_c = |C|$, and let $s = |S|$. Forming the state-specific neural coefficients and performing exact region reduction requires $O(dm)$ arithmetic operations for an m -ridge continuation in dimension d . The screen uses $O(N(1 + m_c))$ fixed-precision interval operations, followed by $O(sm)$ rational operations to compare survivors using the original objective. The vectorized implementation uses $O(N(1 + m_c))$ interval working storage, in addition to the fitted coefficients, and returns $O(s)$ retained indices. These are arithmetic counts; the cost of rational arithmetic also depends on operand bit lengths.*

PROOF: Each ridge requires state and innovation projections and a constant number of exact range tests. Each remaining ridge contributes one scalar interval expression at each lattice point. Pairwise outward summation of the ridge contributions is linear in their number. The exact return evaluates the original m -ridge objective at each survivor. The implementation retains ridge-by-lattice batches, giving the stated storage rather than a streaming bound. Integer index comparison resolves equal rational values. None of these counts eliminates the dependence on d , lattice resolution or fitted width. *Q.E.D.*

Terminal optimization is unchanged in the experiment. The screened intervention concerns only nonterminal trained-ReLU continuations. In particular, time saved by an analytic terminal formula is not credited to learned-feature reduction.

9.3. *Preservation of a prospective economic return*

A solver is used inside a procedure, not in isolation. Consider a deterministic prospective controller conditional on its construction data and random streams. At each attempted resolution it constructs an acquired policy, verifies whole-cell advantages and continuous-action lower covers, and then runs the declared sequence of cost-inference looks. It returns at the first certified economic target, advances resolution when required by its rule, or reports budget exhaustion. Suppose its branching decisions depend on these mathematical outputs, not on elapsed time or solver-specific diagnostic counters.

THEOREM 17—Prospective transcript invariance: *Run the same controller with the algebraic solver and the lossless screened solver. Suppose corresponding construction steps produce identical stored continuations, acquisition partitions and other candidate actions, and use identical inference streams. Then both controllers return the same policy, have identical whole-cell verification endpoints and cost-inference endpoints, and stop at the same resolution and inference look with the same status. Their exact fitted-optimization loss is the same, namely zero on the declared action lattice. Their all-state Bellman-error certificate is unchanged.*

The returned policies have exactly equal expected economic cost under every admissible initial distribution. This equality does not require statistical inference. It does not assert equality of computation cost.

PROOF: Condition on the common construction inputs and streams. Theorem 15, or the exact fallback, gives identical proposed action indices at every leaf and date. The remaining candidate menu and its ordering are the same. The same verifier therefore evaluates the same intervals and resolves the same incumbent-preserving comparisons, producing an identical acquired policy and continuous-action error account. Evaluating that policy on the same interval paths gives identical path endpoints, hence identical inference bounds and the same next controller decision. Induction over inference looks and attempted resolutions proves the entire transcript assertion. No additional attempt occurs after a successful return. The inherited fitted-witness transfer and directed policy-improvement theorems apply

to the identical witnesses and certificates. Finally, identical policies have the same controlled law and expected cost for every initial law. *Q.E.D.*

The theorem is conditional on actual parameter identities, not merely on a shared training seed. The empirical audit checks those identities. It does not apply to a wall-clock deadline rule: a faster solver can reach a different stage before a real-time deadline. Nor does a target that is not attained under the declared finite budgets become impossible in the underlying economic model.

COROLLARY 18—Computation prices at an unchanged policy: *Suppose both implementations return π and face specified computation charges C_A, C_S . At a nonnegative price λ , their total expected policy-plus-computation cost difference is*

$$\{J(\pi) + \lambda C_S\} - \{J(\pi) + \lambda C_A\} = \lambda(C_S - C_A).$$

If an installed controller has a verified replacement-gain interval $[l, u]$, a fixed installation charge τ gives net gain in $[l - \tau - \lambda C_i, u - \tau - \lambda C_i]$ for implementation i .

PROOF: Cancel the identical policy costs in the first display and subtract the specified deterministic charges from the gain interval in the second. *Q.E.D.*

This comparative statement holds for specified resource charges. Three measured process repetitions do not estimate a distribution-free expected runtime, and the study does not calibrate λ or τ . They do, however, permit a directly interpretable work comparison at exactly the same economic and statistical accuracy, without attributing different welfare to identical neural policies.

10. FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

10.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i , let D_i be the difference in expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \overline{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \quad \ell \geq \log(4K/\alpha), \quad (43)$$

where K bounds the number of sampled contrasts.

THEOREM 19—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \quad (44)$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

PROOF: For each drawn bin sequence, conditional integration over its continuous realizations places the exact path expectation between the interval endpoints. Uniform mixing over the equal bins recovers the original continuous law. The lower endpoint expectation is thus no larger than D_i , and the upper endpoint expectation is no smaller. Apply the bounded empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#) to the endpoint samples and take the union bound, with outward arithmetic included in the mean, variance and radius. Conditional validity for every $\mathcal{H}$ implies unconditional validity by integration. On this one simultaneous event, all ensuing deterministic comparisons of the covered quantities hold together. *Q.E.D.*

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

10.2. Replacement-fee identification

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 20—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \quad (45)$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.

PROOF: Subtract the gain intervals from k and compare the resulting interval endpoints with zero. Taking the image of $[l, u]$ under the absolute-value map gives (45). The assertions hold simultaneously for every nonnegative fee on the original coverage event. Q.E.D.

The fee frontier is more informative than a conclusion at one generous charge. The earlier 1/64 replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

10.3. Observation costs on the same bit grid

Let $\mathcal{B} = \{4, \dots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b -bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t < T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 21—Certified choice of actual net implementation cost: *Suppose all $D_b \in [l_b, u_b]$ simultaneously. For any $\lambda \geq 0$ define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \quad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \quad (46)$$

$$\hat{b}(\lambda) = \min \arg \min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\hat{b}} + \lambda s_{\hat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \quad (47)$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\hat{b}(\lambda)$ is weakly decreasing in λ .

PROOF: Subtract the common J_{b_0} . The selected implementation has net difference at most $U(\lambda)$, and the best catalogue implementation has net difference at least $L(\lambda)$. This proves the regret bound pointwise on the one simultaneous event and hence for every λ . A minimum of finitely many affine functions changes its active affine expression only at an intersection. For $\lambda_2 > \lambda_1$, sum the two minimizing inequalities to obtain $(\lambda_2 - \lambda_1)(s_{\hat{b}(\lambda_2)} - s_{\hat{b}(\lambda_1)}) \leq 0$. Since s_b increases strictly with b , the precision comparative static follows. *Q.E.D.*

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (47) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr} , and its actual interpolated critic has modulus L_t^f . Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\text{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \quad (48)$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\hat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (48) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\text{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

11. PROSPECTIVE ECONOMIC STOPPING AND LEARNED STRUCTURE

A policy certificate determines whether a proposed change is safe at every state. An economic stopping rule asks when the returned complete policy has attained a

specified expected-cost target under the initial law. Neither question is answered by choosing the best recorded policy after all computation has finished.

11.1. A first-attainment service

Fix an installed policy π^0 , a target $q \in (0, 1)$, a finite resource ladder, and cumulative sample sizes at each stage. At a stage, construct a critic from primitives, transfer the old policy exactly to any refined partition, and execute a verified simultaneous sweep. Freeze the resulting complete policy before drawing that stage's fresh evaluation stream. At each declared look form a confidence interval for

$$D_{k,q} = \mathbb{E}J_0^{\pi^k} - q\mathbb{E}J_0^{\pi^0}. \quad (49)$$

Return the policy at the first nonpositive upper endpoint. A strictly positive lower endpoint ends the current stage and advances to the next resource stage. An unresolved interval leads to the next declared look. Exhausting the last stage returns the safe incumbent and an explicit budget-exhaustion status. It is not a conclusion that the target is unattainable.

THEOREM 22—Validity at prospective return: *Suppose each accepted sweep satisfies Theorem 8. At each declared stage and look, assume the interval for (49) has conditional failure probability at most $\alpha_{k,l}$ given the information preceding the stage's fresh evaluation stream. If $\sum_{k,l} \alpha_{k,l} \leq \alpha$, then, with probability at least $1 - \alpha$, every successful return satisfies its stated expected-cost target. Every returned incumbent, including one returned after budget exhaustion, is statewise no worse than the original installed policy. No independence between different methods or different looks is required for the union bound. Each within-stage endpoint sample must satisfy its stated conditional sampling assumptions.*

PROOF: Condition on the history preceding each executed stage. Its frozen policy and fresh independent-bin stream satisfy the fixed-policy interval guarantee. Integrating conditional failure probabilities and taking the union over the prespecified potential looks gives a simultaneous event of probability at least $1 - \alpha$. The first-crossing rule selects one already covered interval on this event, so its nonpositive upper endpoint implies (49) is nonpositive. Unexecuted later looks need not

be computed. Statewise safety follows by composition of the deterministic sweep inequalities and exact policy transfer, independently of the economic inference event. Q.E.D.

The theorem also permits a countable predetermined ladder with summable failure allowances. The experiment uses the finite ladder specified below, not an unimplemented anytime extension. A process clock begins before construction and ends after its durable output. Failed stages, own-policy evaluation and inference are charged to that service. A method reaching a target early must not be charged for hypothetical later stages, and a method failing early must not have its completed work omitted.

For the implemented bounded-sample account, each path is enclosed over its complete forty-bit initial and innovation cells. The last innovation is integrated analytically. The interval in Theorem 19 applies at each declared look, using the known support and outward sample moments. The new family limit is 2,048 intervals with logarithm upper bound 14 and error 1/100; the planned maximum is 1,188 intervals. Earlier looks reuse prefixes within a stage and are covered by the same finite union. Distinct stages use fresh streams. A fixed pseudorandom seed establishes reproducibility under the declared independent-bin model, not mathematical independence of a deterministic sequence.

COROLLARY 23—Economic adoption at return: *Suppose the returned policy has cumulative gain $G = \mathbb{E}J_0^{\pi^0} - \mathbb{E}J_0^{\pi} \in [l, u]$ on the simultaneous event. For every replacement charge $\tau \geq 0$, computation price $\lambda \geq 0$, and recorded complete work $W \geq 0$, its net gain belongs to*

$$[l - \tau - \lambda W, u - \tau - \lambda W].$$

Thus $\tau + \lambda W < l$ certifies profitable adoption and $\tau + \lambda W > u$ rules out profitable adoption. These comparisons hold simultaneously over the entire nonnegative price plane on the original event.

PROOF: Subtract the same known quantity $\tau + \lambda W$ from both endpoints of the gain enclosure. The assertion is pathwise on the simultaneous event, so choosing prices after observing recorded work requires no extra confidence allocation. *Q.E.D.*

This is a decision at return, not a claim that the executed relative-cost stopping rule optimizes all possible prices or future computation schedules. The units of λ must convert recorded work into the model's cost units. In an uncalibrated theoretical economy the price plane is an explicit economic sensitivity analysis, not an estimated welfare magnitude.

11.2. Learning an approximately null direction

Suppose the transition is $\mu_t(x, 0) + Ba + sZ$ with unknown $B \in [0, 1]^d$, known nonlinear drift and known bounded additive action-invariant innovation law. Fit a direction $\hat{v}$ on training observations. Independent validation observations Y_i estimate B coordinatewise; each coordinate has mean B_j and range length at most w . Let the recorded mean enclosure be $[\underline{m}_j, \overline{m}_j]$ and let δ_{bin} cover numerical bin displacement. Form

$$\mathcal{K}_N = \prod_{j=1}^d \left[\max\{0, \underline{m}_j - r_N\}, \min\{1, \overline{m}_j + r_N\} \right], \quad r_N = w \sqrt{\ell / (2N)} + \delta_{\text{bin}}. \quad (50)$$

Only validation data, the fitted direction and the declared parameter prior enter the gate.

THEOREM 24—Confidence-qualified action-null correction: *For a finite collection of H validation boxes, each with at most $d_{\max}$ coordinates, $2Hd_{\max}e^{-\ell} \leq \alpha_B$ gives simultaneous coverage of their true exposures with probability at least $1 - \alpha_B$. On this event, for every fitted direction, amplitude M , and nonnegative feasible action pair,*

$$|(P_B^a - P_B^b)M(\hat{v}^\top y + c)_+| \leq |M| \sup_{B' \in \mathcal{K}_N} |\hat{v}^\top B'| |a - b|. \quad (51)$$

A whole-cell critic-advantage upper bound evaluated uniformly over the same box, plus the discounted right-hand side of (51), is a valid upper bound for the corresponding true advantage with the nuisance removed. Accepting only when that bound is nonpositive

preserves safety simultaneously for all tested amplitudes and actions. The simplification is rejected whenever its uncertainty allowance prevents the gate from passing.

PROOF: The bounded-coordinate inequality of [Hoeffding \(1963\)](#) and a union bound give coverage of (50); numerical mean and bin allowances preserve containment. On that event, use a common innovation to couple the two actual kernels. Their next-state difference is $B(a - b)$. The positive-part map is one-Lipschitz, so its projected difference is bounded by $|\hat{v}^\top B| |a - b|$ for every realization. Integration proves (51). The box event controls all linear projections, not merely a preselected direction. Uniformly enclosing the rest of the critic advantage over that same parameter set prevents an unobserved true-parameter oracle from entering the remaining calculation. Subtracting a nuisance difference cannot increase the upper bound by more than its absolute allowance. *Q.E.D.*

The experiment uses $w = 1/2$, $\ell = 14$, 27 validation boxes and at most eight coordinates, giving error at most $1/200$. The direction is fitted on 256 observations and then quantized; the exact stored direction is used in (51). Validation sizes are 512, 4,096 and 32,768. The guarantee concerns an unknown action-exposure vector in a maintained structural model. It does not infer an arbitrary unknown transition law, and coverage failures must remain in the evidence rather than being discarded by reference to the simulator’s known truth.

An empty validation box invalidates the parameter qualification and rejects the simplified certificate. It cannot authorize a policy change.

12. PROSPECTIVE RETURNS IN THE NONLINEAR INVESTMENT ECONOMY

12.1. Design

Tasks are $(d, T) = (2, 2), (4, 4), (8, 6)$. Each method/task/target combination is a separate from-primitives service. Targets $q = 9/10$ and $49/50$ require ten- and two-percent expected-cost reductions relative to installed zero investment. The stopping statistic is the signed contrast $J(\pi) - qJ(0)$, not a ratio of plug-in mean estimates. Two resource stages use respectively 256 and 1,024 training states, 32 and 128 non-tensor leaves, and four and eight innovation bins. Actions are downward

quantized at $1/4096$ using the lower capacity over the complete leaf. Partition geometry is chosen before the generator and is nested between stages.

Trained ReLU, quadratic and ExtraTrees fitted-value methods use their own future targets. The neural architecture and training are described in Section 6. ExtraTrees uses 16 trees, depth at most eight and minimum leaf size three; quadratic FVI fits a ridge-regularized quadratic. The original two-dimensional frontends are reconstructed at $N = 8, 16$, $K = 4$ and $M = 4$. Every method receives nine common robust candidate actions and one fitted proposal, as well as eight intervals covering the full continuous feasible action set. The new algebraic neural action solver is evaluated separately on saved critics; it was not substituted for the frozen experimental construction menu.

After one complete simultaneous sweep at a stage, inference uses cumulative looks of 4,096, 16,384 and 65,536 interval paths. It stops on the first covered target crossing, advances a stage at a strictly positive lower target contrast or exhausted look budget, and never executes a later stage after success. All computation, including failed stages and each method's own inference, is retained. Three isolated sequential processes repeat identical training and inference seeds, with rotated method order, a fixed available CPU core and one numerical-library thread. These measure reproducibility and clock variation, not a population of independently trained networks. Clock frequency is recorded where available but not controlled.

12.2. *The refinement and the complete returned policies*

The first cohort uses the residual/value cache. Its complete 66-process record contains 39 target attainments and 27 budget exhaustions. In the eight-dimensional task it makes no policy change. The separate sensitivity extension was motivated by that completed result and frozen before its own production and fresh inference. It retains the same targets, fitted constructions and common actions, while intersecting the old cache comparison with the signed reference interval of Proposition 10. This chronology matters: the extension is not described as prespecified before observing the primary cohort.

TABLE I

OUTCOMES BEFORE THE SIGNED REFERENCE REFINEMENT

d	T	q	Stages	Paths	Status	Last-pass changes
2	2	49/50	1	16384	Hit	27
2	2	9/10	2	65536	Hit	93
4	4	49/50	2	16384	Hit	90
4	4	9/10	2	16384	Exhausted	90
8	6	49/50	2	65536	Exhausted	0
8	6	9/10	2	4096	Exhausted	0

This primary-cache cohort is not dropped after its limitations become visible. These are the neural rows; the candidate generators return policy-identical final arrays within each primary task and target. Exhausted means the declared finite service did not certify its target, not that improvement is impossible. All own construction, inference and failed-stage work remains charged.

TABLE II

ACTUAL COSTS AT THE PROSPECTIVE NEURAL-POLICY RETURN

d	T	q	Paths	Expected cost	Installed minus returned
2	2	49/50	4096	[0.5338, 0.6834]	[0.0622, 0.2408]
2	2	9/10	16384	[0.5950, 0.6503]	[0.1294, 0.1777]
4	4	49/50	4096	[0.8975, 1.0658]	[0.1199, 0.3796]
4	4	9/10	4096	[0.8974, 1.0658]	[0.1177, 0.3775]
8	6	49/50	4096	[1.3427, 1.5356]	[0.0918, 0.4154]
8	6	9/10	16384	[1.4216, 1.4788]	[0.2145, 0.2978]

N is the trained ReLU generator. Every row is a separately executed target service with the signed reference certificate. Endpoints are rounded outward. The interval covers the complete implemented policy under the original continuous law. Timing repetitions have identical policy and inference identities and are not additional economic observations. Other generators and every earlier-stage outcome are retained in the supplement.

All 66 signed-reference services attain their targets at the first resource stage. This means one complete all-date sweep on 32 leaves, not T exact sweeps and not attainment of the Bellman optimum. For the neural eight-dimensional, six-date controller, the changed leaf counts by date are (26, 24, 24, 20, 14, 0). Its identified

cost reduction is generated by nonterminal decisions; it is not a terminal-repair experiment relabeled as a long-horizon result. At the ten-percent target its cumulative gain interval is $[0.21452658, 0.29774911]$, with endpoints rounded outward. The complete expected cost interval is $[1.42162445, 1.47873990]$.

By Corollary 23, the eight-dimensional return supports every replacement-and-computation charge below its lower gain endpoint, in the model’s normalized units. A charge above its upper gain endpoint rules out profitable adoption. The interval between them remains unresolved. This translates the returned policy into a full economic price sensitivity, rather than choosing a favorable fee after inspecting one estimate.

TABLE III

WHERE THE SIGNED REFERENCE CERTIFICATE CHANGES DECISIONS

Date	Changed	Base width	Refined width	$U - C$	$C - L$	$U - L$
0	26	14.094	0.610	0.604	0.418	1.021
1	24	12.158	0.594	0.588	0.407	0.994
2	24	10.094	0.569	0.562	0.388	0.950
3	20	7.892	0.530	0.521	0.358	0.879
4	14	5.543	0.466	0.453	0.310	0.762
5	0	1.055	1.055	0.640	0.074	0.713

Eight-dimensional state, six dates, trained ReLU, first stage, 32 non-tensor leaves. Widths are maxima over candidate enclosures; decomposition entries are maxima over leaves, rounded upward. The last row uses analytic terminal differences rather than differentiating an acquired actor. A tighter comparison certificate is not itself a new policy-cost observation.

The directed comparison, all-state optimality account and initial-distribution cost interval continue to answer different questions. At date zero of the eight-dimensional neural service, the maximum candidate interval width falls from more than 14 to less than 0.610 after the signed refinement. The full-action greedy allowance remains about 1.021, and the inherited finite-sweep all-state bound is much looser than the observed expected cost gain. The experiment does not claim that 32 leaves or one sweep solve the continuous-action Bellman problem accurately at every state. It does establish the declared prospective expected-cost

improvement for the complete implemented policy, together with deterministic statewise incumbent protection.

12.3. Method comparisons and complete work

TABLE IV
COMPLETE PROCESS WORK TO THE TEN-PERCENT COST TARGET

d	T	Method	Min (s)	Median (s)	Max (s)	Output KiB	RSS MiB
2	2	W	0.781	0.808	1.345	739	42.0
2	2	E	1.678	1.737	1.740	1100	110.0
2	2	Q	0.695	0.701	0.745	781	42.3
2	2	N	1.389	1.409	1.431	788	50.6
2	2	F	0.807	0.809	0.853	731	42.0
4	4	E	11.515	11.684	11.781	1286	107.3
4	4	Q	5.576	5.582	5.583	489	39.0
4	4	N	9.073	9.087	9.238	512	50.7
8	6	E	31.833	31.886	35.303	2588	112.3
8	6	Q	21.027	21.243	21.882	1350	43.8
8	6	N	26.714	26.771	30.501	1400	51.8

N: trained ReLU; Q: quadratic FVI; E: ExtraTrees FVI; W: compiled witness; F: tensor FVI. Three isolated sequential processes use identical seeds within a method. Each clock includes its own imports, construction, verification, inference and durable output. Compare methods within a task runner, not across runners. Affinity and numerical-library threads are fixed; clock frequency is not controlled. Output excludes prior-stage archives outside the service; RSS is process peak memory. Both targets and all repetitions appear in the supplement.

The trained ReLU proposal changes no deployed cell beyond the nine common actions in either cohort. The only tenth-proposal effect is one compiled-witness cell in the two-dimensional sensitivity task, repeated across targets and timing runs. These identities are checked explicitly rather than inferred from similar cost intervals.

The conventional generators receive the same acquired states, common candidate actions, signed reference information and economic stopping rule. The audit

reconstructs the trained proposal and then removes the tenth candidate from the recorded gate to determine whether that proposal changes the selected action. This is a deterministic candidate-attribution diagnostic, not a fresh policy-cost experiment. Exact deployed-policy identities are checked across generators and across timing repetitions.

In the four- and eight-dimensional sensitivity tasks, the generators return policy-identical arrays within each target. In the two-dimensional task the compiled-witness return differs slightly, while the trained ReLU, quadratic, ExtraTrees and tensor-FVI returns coincide. Thus the policy improvement is not evidence that the neural generator dominates these alternatives. At the ten-percent target, quadratic FVI has the lowest recorded median complete-process work in each task. The neural backend supplies a genuinely trained nonlinear continuation and analytic integration, but those facilities must not be confused with an identified economic ranking.

The clocks include imports, primitive construction, critic fitting, acquired-policy preparation, verification, own inference, serialization and durable outputs. Method comparisons are within each task’s controlled runner; clocks from different task runners are not a hardware-controlled scaling regression.

12.4. Neural action witnesses and learned null directions

TABLE V

EXACT LATTICE SEARCH ON SAVED FITTED NEURAL CRITICS

d	T	Full lattice	Candidates	Old index	Exact index	Critic decrease
2	2	896	10	279	303	0.0000800
4	4	768	4	191	169	0.0000289
8	6	640	6	559	555	0.0000012

One declared cell center and its saved next-date critic per task. Rational isolation and comparison agree with exhaustive rational lattice minimization. The reported decrease is in that fixed approximate Bellman objective, not actual economic policy cost. These solver regressions neither retrain a critic nor produce new policy-cost observations.

The algebraic action solver gives exact fitted-objective minimizers on the full deployed lattice in its retained regressions, using only a small subset of lattice indices. This is a constructive neural-search result. Its objective decreases concern the saved approximate continuation at a specified cell center, not the expected cost of a new deployed controller. All historical inference remains attached to the policies that were actually evaluated.

TABLE VI
VALIDATION-QUALIFIED LEARNED DIRECTIONS

d	Validation rows	Projection-bound range	Corrected changes	Harmful plug-in
2	512	[0.0660, 0.0670]	36	0
2	4096	[0.0241, 0.0271]	54	0
2	32768	[0.0121, 0.0141]	75	0
4	512	[0.0608, 0.0636]	0	0
4	4096	[0.0217, 0.0232]	0	0
4	32768	[0.0081, 0.0102]	0	0
8	512	[0.0586, 0.0622]	0	0
8	4096	[0.0207, 0.0236]	0	0
8	32768	[0.0078, 0.0101]	0	0

Each row aggregates three independently fitted directions and three amplitudes on 64 leaves per case.

The projection range reports the minimum and maximum robust upper allowance across fitted directions, not a confidence interval for a population mean. All 27 exposure boxes cover the simulator exposure in this record. No strictly harmful plug-in choice is identified here; the earlier known-perturbation catalogue is a different diagnostic and remains unchanged.

The sample-split study fits nine directions across dimensions two, four and eight and three independent direction-fitting seeds. Three validation sizes give 27 confidence boxes; three amplitudes on 64 acquired leaves give 81 cases and 5,184 cell-case decisions. All 27 recorded boxes contain the simulator exposure. The corrected gate makes 165 safe changes. No strictly harmful plug-in decision is identified in this learned-direction catalogue. That zero count is retained; it is not replaced by the large harmful count from the earlier, different known-perturbation experiment.

The primary policy-cost, sensitivity policy-cost and learned-exposure families have errors $1/100$, $1/100$ and $1/200$, respectively. Their joint union error is at most $1/40$. This addition combines covered statements; it does not make identical policies, timing repetitions, amplitudes or reused prefixes independent observations. The learned-direction conclusion remains conditional on the known drift and maintained additive action-invariant innovation structure.

12.5. *Relation to the original evidence*

The earlier witness, FVI, acquisition, common-repair, exact-null and adaptive-grid results remain in the complete development article and its supplement. In particular, earlier adverse witness-cost comparisons are not erased by the improvement of a newly installed zero controller. The new first-attainment services replace retrospective catalogue work as the primary work-to-target endpoint for this experiment; the historical catalogue remains a valid account of a different executed procedure. The full source and result archive preserves both and states which economic policy and law each interval actually covers.

13. EXECUTING FITTED NEURAL ACTION WITNESSES

The preceding study establishes a prospective economic return and a non-tensor verification procedure. Its candidate-attribution audit also identifies a necessary next comparison: the common action menu often determines the same policy regardless of the fitted generator. The algebraic solver in Theorem 7 was tested there as a solver, not used to create the policies whose costs were reported. We now execute that solver in new prospective services and evaluate the resulting policies on new streams. The original investment laws, Bellman optimum and NBO research object are unchanged.

13.1. *A controlled action-search comparison*

The new protocol fixes the three tasks $(d, T) = (2, 2), (4, 4), (8, 6)$ before production. It compares five modes. Common-only uses the nine robust actions without fitting a critic. ReLU-menu uses a genuinely fitted 32-feature continuation and adds its 17-menu proposal. ReLU-exact retains every one of those candidates and adds

the exact fitted-critic lattice minimizer. Quadratic-exact supplies the analogous exact action search for its fitted quadratic continuation. ExtraTrees-menu supplies its own future tree-based proposal. All modes receive the same exact terminal-action search, so an analytic terminal improvement is not assigned to neural representation.

The nonterminal verification rule is identical across generators: signed sensitivity of the installed zero-investment policy from Proposition 10. Its state boxes, innovation bins and primitive arithmetic do not depend on the fitted critic. Shared candidate actions therefore have identical upper endpoints. This design implements the premise of Corollary 13, rather than changing the candidate generator and its error certificate simultaneously.

Each mode is reconstructed independently for each task, target and one of three distinct training seeds. ReLU-menu and ReLU-exact use the same seed within an attempt, but neither loads the other's fit free of charge. Their fitted-parameter identities are checked. Different seeds produce different fitted neural objects. The three seeds are not a population of economic tasks, and their timing variation is not described as a controlled timing-only repetition. The preceding fixed-seed timing repetitions remain a separate experiment.

13.2. Prospective economic targets

The two targets require ten- and twenty-percent expected-cost reductions from the installed zero policy. Each target service starts from primitives. It has two declared construction attempts, with training rows 256, 1024, acquired leaves 32, 64, and innovation bins 4, 8. The same locally split non-tensor observation geometry and downward action spacing $1/4096$ apply to every mode.

At each attempt a complete candidate is certified against the installed policy and frozen before inference. Intermediate attempts are not deployed. In particular, refinement does not apply the smooth-reference derivative formula to an already changed discontinuous actor. Every attempt is safe relative to the installed zero policy; no ordering between the policies of different attempts is assumed.

The own-service stopping estimand is $J(\pi) - qJ(0)$, with $q = 9/10$ or $4/5$, rather than a ratio of two estimated means. Its cumulative inference looks have 4096, 16384

and 65536 paths. A nonpositive upper endpoint returns immediately. A positive lower endpoint advances the construction attempt; an unresolved interval advances the declared look. After the final unsuccessful attempt the service returns a budget-exhausted safe policy, not a claim that the target is impossible. Every attempted fit, exact search, verification, inference and durable output belongs to the service's surrounding process clock.

There are 90 separately executed target services. After each set of five modes has returned, a separate 65,536-path stream compares ReLU-exact with each other returned policy. This stream is independent of construction and stopping information under the declared independent-bin model. It evaluates the original continuous initial and innovation laws using forty-bit bin enclosures and analytic final integration. Negative path differences are retained. Full policy identities receive exact zero contrasts; other overlapping intervals are not treated as identities.

The new finite confidence family has error at most $1/100$. Its maximum possible interval count is

$$90 \cdot 2 \cdot 3 \cdot 3 + 18 \cdot 4 = 1692 < 2048,$$

covering own cost, target contrast and gain at every possible look, together with the independent validation contrasts. The retained logarithm allowance is checked by exact rational arithmetic. The first-attainment argument of Theorem 22 applies conditionally at every fresh stage. This finite-look account is not an undeclared confidence sequence valid at arbitrary future sample sizes.

The new cohort executes 90 separate prospective target services. Of these, 75 attain their declared target and 15 return a budget-exhausted safe candidate. The audit regenerates 78 distinct fitted-model certificate and action-search calculations and checks 330 stopping looks. In the trained ReLU-exact services, 1354 nonterminal cell-attempt records have strictly positive menu optimization loss, while the additional algebraic candidate changes 9 deployed cell-attempt decisions. These counts refer to the declared service records, not independent samples. Among the 72 fresh paired comparisons with ReLU-exact, 0 identify lower ReLU-exact cost, 0 identify higher cost, 42 are exact policy identities and 30 remain unresolved.

TABLE VII
PROSPECTIVE TWENTY-PERCENT TARGET SERVICES

d/T	Mode	Reached	Cost upper	Median (s)	Range (s)
2/2	C	3/3	0.61949	3.776	3.763–3.793
2/2	N	3/3	0.61949	4.511	4.497–4.536
2/2	X	3/3	0.61948	5.216	5.179–5.238
2/2	Q	3/3	0.61949	4.083	4.043–4.107
2/2	E	3/3	0.61962	5.185	5.168–5.202
4/4	C	3/3	0.97342	12.881	12.439–12.983
4/4	N	3/3	0.97342	15.077	15.004–15.240
4/4	X	3/3	0.97342	17.951	17.786–18.369
4/4	Q	3/3	0.97342	13.780	13.475–13.991
4/4	E	3/3	0.97342	15.984	15.872–16.513
8/6	C	0/3	1.36544	35.153	35.049–35.571
8/6	N	0/3	1.36544	39.239	39.190–39.458
8/6	X	0/3	1.36544	44.189	44.183–45.418
8/6	Q	0/3	1.36544	38.682	38.257–38.954
8/6	E	0/3	1.36544	40.766	40.704–41.016

C uses common candidates without fitting. N uses the trained ReLU menu; X adds its exact algebraic witness; Q uses quadratic exact search; E uses ExtraTrees menu proposals. All modes receive the same exact terminal action. Three different training seeds are reconstructed separately. Cost upper is the largest returned-policy upper endpoint across those seeds, not a confidence bound for a population mean. Complete process times include every own failed attempt and inference look.

13.3. What the neural action computation changes

Table VIII separates positive center-objective optimization loss, strict improvement of a whole-cell upper endpoint and an actual change of the deployed action. These events are not interchangeable. The center optimization loss is evaluated exactly from the same stored critic; it is zero for the algebraic minimizer. Removing only the added algebraic candidate reconstructs the within-attempt menu policy with the same verifier. Thus every reported stricter endpoint is a genuine nested-menu improvement.

TABLE VIII
OBJECTIVE SEARCH AND WHOLE-CELL DEPLOYMENT ATTRIBUTION

d	Mode	$\kappa > 0$	Changed	Stricter	Roots	Values
2	X	163	7	7	337	3632
2	Q	145	0	0	321	2276
4	X	451	2	2	626	10849
4	Q	224	0	0	409	3965
8	X	740	0	0	840	14603
8	Q	37	0	0	127	4866

Counts are cell-attempt records across all own target services, including repeated reconstruction at the other target; they are not independent economic observations. Positive optimization loss refers to the nonterminal menu witness compared with the exact lattice witness. Changed and stricter counts remove only the extra algebraic proposal while retaining the shared verifier. Root and value counts include the common terminal solver, which is not credited as a neural-specific contribution.

The new execution supplies the previously missing connection between the trained representation’s exact action witness and a deployed policy. It also measures the extent of that connection. A large number of positive optimization losses can coexist with few additional deployed actions because the common safe menu already determines most selected upper bounds. Even a strictly improved upper bound does not prove that the resulting policy costs less than its menu counterpart. The independent paired contrasts in Table IX, and not the fitted objective or interval width, determine that economic comparison. Every individual interval is reported in the supplement.

The common-only control pays no fitting cost and is a substantive comparator. Quadratic-exact receives the same algebraic-search opportunity, while ExtraTrees retains its declared fitted-value construction. This prevents the exact solver’s availability from being conflated with a special verification privilege. The experiment compares the complete returned procedures and identifies their components; it does not presume that fitting a neural continuation must beat a common-action or conventional-value alternative.

TABLE IX
FRESH RETURNED-POLICY COST COMPARISONS

d	X versus	X lower	X higher	Identity	Unresolved
2	C	0	0	2	4
2	N	0	0	2	4
2	Q	0	0	2	4
2	E	0	0	2	4
4	C	0	0	3	3
4	N	0	0	4	2
4	Q	0	0	3	3
4	E	0	0	2	4
8	C	0	0	6	0
8	N	0	0	6	0
8	Q	0	0	6	0
8	E	0	0	4	2

Each row covers both targets and all three training seeds. A negative paired difference favors X. Identity means equal complete policy arrays on the same observation partition, not merely overlapping intervals. Other unresolved intervals are not equivalence findings. The fresh comparison stream is independent of the corresponding stopping information under the sampling model.

13.4. *Work, uniform accuracy and adoption*

The main table reports complete own-target process work at a matched economic target. It includes failed intermediate attempts and all inference actually needed to return. The supplementary work ledger separately records fitting, root isolation, candidate evaluation, signed verification, path evaluation, serialized bytes and resident memory. Different tasks run on different task runners; within each task processes are sequential, use one available core and one numerical-library thread, and rotate mode order. Available governor information is recorded, but clock frequency is not controlled.

The fresh paired validation is additional study work. Its recorded timer ends after endpoint evaluation and serialization, before the final comparison JSON is committed. It is therefore reported as a partial validation-service clock, not a com-

plete process clock, and is not concealed inside the own-target time. The protocol's intended complete comparison-process timing was not implemented; this timing boundary does not change the prospective stopping rule or its fully recorded own-service work. The raw timing records are retained without retrospective adjustment.

Every returned policy also receives a one-sweep all-state loss account. At date t it includes the old-reference continuation term $\beta_t E_{t+1}^0$ as well as the directed greedy allowance. The tables use a valid support bound for that reference error and cap the result by the original total-cost support. They do not substitute the local $U - L$ gap for loss against the Bellman optimum. A successful expected-cost target under the initial law is thus not mislabeled as exact Bellman convergence.

The gain from replacing installed zero investment remains an actual economic gain, subject to the stated confidence event and normalized primitives. Choosing between two returned methods requires the additional incremental-cost calculation (36). It charges the measured work difference and any replacement fee, and uses the fresh paired policy-cost interval. The complete fee-computation-price regions follow from that interval without choosing a favorable fee after seeing the outcome. The experiment is a theoretical investment exercise, not an externally calibrated welfare estimate.

14. EQUAL-POLICY WORK AT PROSPECTIVE ECONOMIC TARGETS

The preceding experiment asks whether an exact trained-neural witness changes a returned policy. A different, necessary question is how economically that same witness can be computed. The lossless screen permits a controlled comparison in which economic accuracy, fitted parameters, whole-cell verification and statistical stopping remain identical. We integrate the separately frozen R58 execution into this revision and independently reconstruct its solver, certificate and path calculations. The integration creates no new training run or independent statistical observation.

14.1. *A matched intervention within Neural Bellman Operators*

The experiment retains the original nonlinear cyclic investment laws. Its tasks are $(d, T) = (2, 2), (4, 4), (8, 6)$ and its targets are ten- and twenty-percent expected-cost reductions relative to installed zero investment. Each task-target has two solver

1 modes and three sequential complete-process repetitions, for 36 services. Both 1
 2 modes independently train the same 32-feature own-future ReLU continuation. 2
 3 Exact stored-parameter hashes establish matching; the common seed alone is not 3
 4 used as proof of matching. 4

5 The sole intervention is the nonterminal fitted-action solver. Algebraic search 5
 6 partitions the hinge objective and isolates its cubic stationary roots. Screened search 6
 7 performs exact activation-region reduction, outward lattice elimination and rational 7
 8 comparison of survivors. Both use the same robust action spacing $1/4096$, the same 8
 9 smallest-index tie rule and the same exact terminal solver. They retain the nine 9
 10 common actions and the critic-menu proposal; the exact action is an additional 10
 11 candidate. The same signed installed-reference verifier supplies all whole-cell upper 11
 12 endpoints and continuous-action lower covers. 12

13 The acquired observation geometry is the existing locally split binary partition, 13
 14 not a tensor grid. The two resource attempts use $(256, 32, 4)$ and $(1024, 64, 8)$ for 14
 15 training rows, acquired leaves and innovation bins. All attempts are checked against 15
 16 installed zero investment. Intermediate candidates are not deployed, and no deriva- 16
 17 tive argument is applied to a changed discontinuous acquired actor. This experiment 17
 18 is a prospective refinement service, not a claim that two refinement attempts are 18
 19 two policy-iteration sweeps. 19

21 14.2. First attainment, including nonattainment 21

22
 23 At each attempted resolution the complete policy is frozen before inference. The 23
 24 target estimand is 24

$$26 \quad D_q = J(\pi) - qJ(0), \quad q \in \{9/10, 4/5\}. \quad 26$$

27
 28 Cumulative sample sizes are 4096, 16384 and 65536. A nonpositive upper endpoint 28
 29 of D_q returns immediately. A strictly positive lower endpoint advances the construc- 29
 30 tion attempt. An unresolved interval advances the inference look; an exhausted 30
 31 look budget advances the attempt. Failure to certify after the last attempt returns 31
 32 a safe policy with status *budget exhausted*. The protocol specifies these rules before 32

its production execution, rather than selecting a favorable pass from a completed catalogue.

The R58 stopping stream is distinct from the R57 stopping and comparison streams. Within a task, target and attempt, the two solver modes and timing repetitions deliberately regenerate the same stream. Repeated computation of that stream does not multiply sample size. The confidence family has error at most 1/100 under the maintained independent-bin model. Its conservative maximum possible number of interval evaluations is

$$36 \cdot 2 \cdot 3 \cdot 3 = 648 < 2048,$$

including cost, target contrast and gain at every possible look. The frozen empirical-Bernstein implementation retains its verified logarithm allowance. The union bound requires no independence between policies or looks. It does not authorize unplanned sample sizes or an arbitrary future stopping rule.

TABLE X
COMPLETE SERVICE WORK FOR EXACTLY MATCHING POLICIES

(d, T)	Reduction	Return	Algebraic (s)	Screened (s)	Ratio
(2,2)	10%	Met	1.212	1.101	1.101
(2,2)	20%	Met	3.577	3.409	1.049
(4,4)	10%	Met	5.837	5.487	1.064
(4,4)	20%	Met	20.554	19.373	1.061
(8,6)	10%	Met	20.229	19.270	1.050
(8,6)	20%	Exhausted	63.827	62.075	1.028

Each time is the median of all three separately reconstructed fixed-work process repetitions. The ratio is algebraic divided by screened. Corresponding policies, fitted parameters, verification endpoints and stopping histories agree exactly. Exhaustion is not successful target attainment. Different tasks use different runners; the comparison is within task and target.

Thirty services attain their target. Six exhaust their budgets: the two modes, each repeated three times, at $(d, T) = (8, 6)$ and the twenty-percent target. Their final target-contrast interval contains zero. The returned policy nevertheless has

TABLE XI
ORIGINAL-LAW ECONOMIC COSTS AT THE COMMON RETURN

(d, T)	Reduction	Policy cost	Gain from installed	Target contrast
(2,2)	10%	[0.5964, 0.6517]	[0.1302, 0.1786]	[-0.0983, -0.0548]
(2,2)	20%	[0.5984, 0.6203]	[0.1595, 0.1740]	[-0.0171, -0.0060]
(4,4)	10%	[0.8995, 1.0672]	[0.1205, 0.3800]	[-0.2482, -0.0056]
(4,4)	20%	[0.9527, 0.9718]	[0.2616, 0.2809]	[-0.0324, -0.0167]
(8,6)	10%	[1.4177, 1.4748]	[0.2130, 0.2963]	[-0.1235, -0.0456]
(8,6)	20%	[1.3464, 1.3649]	[0.3334, 0.3558]	[-0.0141, 0.0050]

The two modes have identical intervals, not independent estimates. Endpoints are rounded outward.

The signed target contrast, rather than a ratio of estimated means, determines attainment. The twenty-percent target in eight dimensions remains unresolved at the final budget.

a positive gain interval relative to installation. Thus failure to certify the stronger target is not failure of policy safety, evidence of zero gain, or a proof that the economic target is impossible.

Every matched pair has identical fitted parameters, acquired actions, certificates, original-law path endpoints and stopping decisions. Consequently its policy-cost difference is exactly zero under every initial law, by Theorem 17. No separate paired statistical experiment is needed to establish this equality. In contrast, the distinct neural and conventional policies in Section 13 retain their own paired cost intervals and must not be assigned this identity conclusion.

14.3. What the computational intervention removes

The machine-independent records identify the mechanism. Across the frozen workloads, the screen removes noncrossing integrated ridges exactly and retains one lattice point per nonterminal neural query. That point is compared rationally against the original objective. No production call uses the algebraic fallback. The number of original-objective rational evaluations therefore falls to the number of neural queries, whereas the algebraic mode compares several candidate points per query and performs cubic-root isolation. The table charges all lattice points evaluated by the interval screen; it does not count only survivors as total work.

TABLE XII

NONTERMINAL NEURAL SEARCH WORK, INCLUDING REJECTED LATTICE POINTS

(d, T)	Reduction	Queries	Exact A	Exact S	Roots A	Screened points
(2,2)	10%	32	189	32	12	23040
(2,2)	20%	96	612	96	43	70144
(4,4)	10%	96	723	96	41	62976
(4,4)	20%	288	2283	288	112	192000
(8,6)	10%	160	1033	160	27	94720
(8,6)	20%	480	3467	480	126	289280

A denotes algebraic search and S screened search. Counts sum all attempted resolutions in one service; all three repetitions reproduce them. The screened mode retains one point per nonterminal neural query and performs zero root isolations and zero fallbacks in this catalogue. Terminal work is common and excluded from these neural counts.

The screen is not an approximate optimizer with a favorable tolerance. The audit recomputes both solvers from every distinct saved model, compares every recorded witness and endpoint hash, regenerates each whole-cell certificate, and reconstructs the original interval paths. Distinct computations are memoized only after their input identities are checked. Every service file and stopping transcript is still compared. Separate exact fixtures verify all lattice endpoints for disjoint small problems, signed weights, activation boundaries, ties, non-dyadic lattices and fallback cases.

Median action-search times decrease in all six task–target groups, as do the complete process medians. The full-service algebraic-to-screened ratios range from approximately 1.028 to 1.101. These gains are smaller than the action-search gains because training, common terminal search, verification and inference are unchanged. The eight-dimensional unsuccessful target is included in the work comparison: faster exhaustion is not described as faster successful attainment.

The effect is a measured implementation gain within the trained NBO procedure at exactly matching policies and statistical accuracy. It is not a ranking of neural methods above common-only, quadratic or tree-based alternatives. The stronger

conventional comparisons, the common-menu attribution results and all unresolved differences in the preceding studies remain part of the evidence.

14.4. *Complete work and the economic decision*

Every displayed service clock surrounds a separate process from invocation through exit and log flush. It includes imports, source checks, independently repeated fitting, action search, signed verification, own path evaluation and inference, all attempted resolutions and durable records. Available frequency-governor information, processor affinity, numerical-library versions, output bytes and peak resident memory are retained. The parent driver's own checkpoint-writing overhead is outside that measured interval and is identified as controller overhead, not assigned a zero cost.

Within a task, processes run sequentially on one available core with one numerical-library thread, and solver order rotates by repetition and target. All three measurements are printed in the supplement. Frequencies are observed where available but not controlled. Different task runners do not supply a hardware-controlled dimension experiment. Three fixed-work repetitions establish neither a distribution-free runtime advantage nor a population of independently fitted policies. The three distinct fitted seeds in the preceding experiment answer a different question and remain separately identified.

For a specified computation price, Corollary 18 converts an equal-policy resource difference directly into a total-cost difference. Relative to the installed controller, the same actual gain interval supports a larger region of installation fees and computation prices when the specified implementation charge is lower. This is a substantive difference in the cost of obtaining an unchanged economic answer, not a claim that two representations of the same policy produce different welfare. The model's units remain normalized; no empirical installation fee, computation price or welfare calibration is inferred from the timings.

The original accuracy objective remains visible throughout. Every returned policy retains its one-sweep all-state loss account, including the installed-reference continuation bound as well as its full-action directed gap. Lossless search changes neither

that account nor the amount of acquisition information. A successful initial-law cost-reduction target is not substituted for convergence to the original Bellman optimum. The constructive resource theory, fitted-witness transfer and directed finite-sweep theorem continue to state the conditions under which that stronger objective is controlled.

15. CONCLUSION

Neural Bellman Operators connect an own-future continuation to a feasible action and an economic cost account. The constructive backend retains the original Bellman-accuracy objective. The trained backend now has analytic innovation integration and a terminating exact action-lattice search, while its proposals remain subject to whole-cell true-advantage verification. These are complementary parts of the same numerical method, not interchangeable claims about the name of a function class.

The non-tensor cache and signed reference calculation turn the policy account into an executable prospective service. Their proofs retain the cost of leaf membership, critic evaluation, state dimension and horizon. The declared expected-cost targets are attained in the sensitivity experiment, with genuine nonterminal action changes and complete own-service clocks. Conventional generators share the same verification treatment, and policy identities make their role observable. The finite experiments identify successful returns and their costs; they do not establish universal neural dominance or calibrated welfare effects.

The full-action gap, the actual expected cost and the price of acquiring or verifying a policy remain separate economic quantities. Keeping all three explicit permits a decision maker to choose the appropriate contract without treating fitting accuracy, certificate sharpness or a favorable timing observation as a substitute for the cost of the implemented policy.

The new action-search execution completes another part of that chain. Exact rational witnesses eliminate the stored critic's lattice optimization loss, and the transfer theorem identifies the remaining acquisition, evaluation, continuous-cover and verification obligations. The nested-candidate comparison makes a stricter certified upper endpoint attributable to an added witness under a fixed verifier.

New prospective services and independent returned-policy contrasts determine the actual economic consequences. Neither exact fitted minimization nor a tighter upper endpoint is substituted for those contrasts. The complete records retain all modes, seeds, targets, attempted refinements and computational costs.

The lossless-search result makes that distinction operational at an unchanged answer. Exact activation-region reduction and outward screening preserve the trained-neural witness; conditional transcript invariance preserves the returned policy, all-state certificate and first-attainment rule. The separately frozen matched execution records lower complete-process median work in each declared task–target group, while retaining the unsuccessful target as unsuccessful. Its equal-policy resource comparison is distinct from the earlier comparisons between different neural and conventional policies. The original Bellman objective, feasibility requirements and economic laws are unchanged.

APPENDIX A: DEVELOPMENT AND REPRODUCIBILITY MAP

The complete development article and proof companion preserve every preceding theorem, economic application and numerical comparison, including adverse results. The prior active article and supplement are also retained as separate development documents. The present main article reorganizes the original constructive core around trained action witnesses, non-tensor verification and prospective economic stopping; it does not replace the NBO subject with a different model.

The publication audit regenerates all unique recorded certificates from their saved fitted critics, replays all stopping decisions and verifies original moment bounds against exact rational sample moments. New solver regressions compare rational action witnesses with exhaustive lattice minimization. These operations do not create new policy-cost samples or replace historical service clocks. The source freezes, ordinary manuscript sources, complete service registry, raw arrays, response and release manifests are included in the revision directory. Mathematical peer review remains distinct from executable regression and document reproduction.

APPENDIX: REFERENCES

- ERNST, DAMIEN, PIERRE GEURTS, AND LOUIS WEHENKEL (2005): “Tree-Based Batch Mode Reinforcement Learning,” *Journal of Machine Learning Research*, 6, 503–556. [3, 16]
- FELZENSZWALB, PEDRO F. AND DANIEL P. HUTTENLOCHER (2012): “Distance Transforms of Sampled Functions,” *Theory of Computing*, 8 (19), 415–428. [9]
- FERNS, N., P. PANANGADEN, AND D. PRECUP (2004): “Metrics for Finite Markov Decision Processes,” in *Proceedings of the Twentieth Conference on Uncertainty in Artificial Intelligence*, 162–169. [4]
- GRIMM, C., A. BARRETO, S. SINGH, AND D. SILVER (2020): “The Value Equivalence Principle for Model-Based Reinforcement Learning,” in *Advances in Neural Information Processing Systems*, 33. [4]
- HOEFFDING, WASSILY (1963): “Probability Inequalities for Sums of Bounded Random Variables,” *Journal of the American Statistical Association*, 58 (301), 13–30. [44]
- KAKADE, S., AND J. LANGFORD (2002): “Approximately Optimal Approximate Reinforcement Learning,” in *Proceedings of the Nineteenth International Conference on Machine Learning*, 267–274. [3]
- MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): “Empirical Bernstein Bounds and Sample Variance Penalization,” in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [38]
- MC SHANE, EDWARD J. (1934): “Extension of Range of Functions,” *Bulletin of the American Mathematical Society*, 40 (12), 837–842. [8]
- METELLI, ALBERTO MARIA, MATTEO PIROTTA, DANIELE CALANDRIELLO, AND MARCELLO RESTELLI (2021): “Safe Policy Iteration: A Monotonically Improving Approximate Policy Iteration Approach,” *Journal of Machine Learning Research*, 22 (97), 1–83. [3]
- MUNOS, R. (2003): “Error Bounds for Approximate Policy Iteration,” in *Proceedings of the Twentieth International Conference on Machine Learning*, 560–567. [3]
- MUNOS, RÉMI, AND CSABA SZEPESVÁRI (2008): “Finite-Time Bounds for Fitted Value Iteration,” *Journal of Machine Learning Research*, 9 (27), 815–857. [3, 16]
- NG, A. Y., D. HARADA, AND S. RUSSELL (1999): “Policy Invariance under Reward Transformations: Theory and Application to Reward Shaping,” in *Proceedings of the Sixteenth International Conference on Machine Learning*, 278–287. [4]
- PIROTTA, MATTEO, MARCELLO RESTELLI, ALESSIO PECORINO, AND DANIELE CALANDRIELLO (2013): “Safe Policy Iteration,” in *Proceedings of the 30th International Conference on Machine Learning*, vol. 28 of *Proceedings of Machine Learning Research*, 307–315. [3]
- SCHERRER, BRUNO, MOHAMMAD GHAVAMZADEH, VICTOR GABILLON, BORIS LESNER, AND MATTHIEU GEIST (2015): “Approximate Modified Policy Iteration and its Application to the Game of Tetris,” *Journal of Machine Learning Research*, 16 (49), 1629–1676. [3]